Introduction to Biomaterials

Lecture [2]

Basics of Material Science, Properties, Testing, and Failures

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Goals

After reading this chapter the student will understand the following:

- Basic material sciences definitions
- The mechanical properties and behavior of materials.
- Material failure under ductile or brittle conditions.
- The time-dependent mechanical behavior of materials.
- Corrosion and its various forms.
- Concepts related to the basic surface properties of materials.

Everyday vs Engineering Thinking

- In daily life, we describe materials as:
  - Soft / Hard
  - Flexible / Rigid
  - Strong / Weak
  - Tough / Brittle

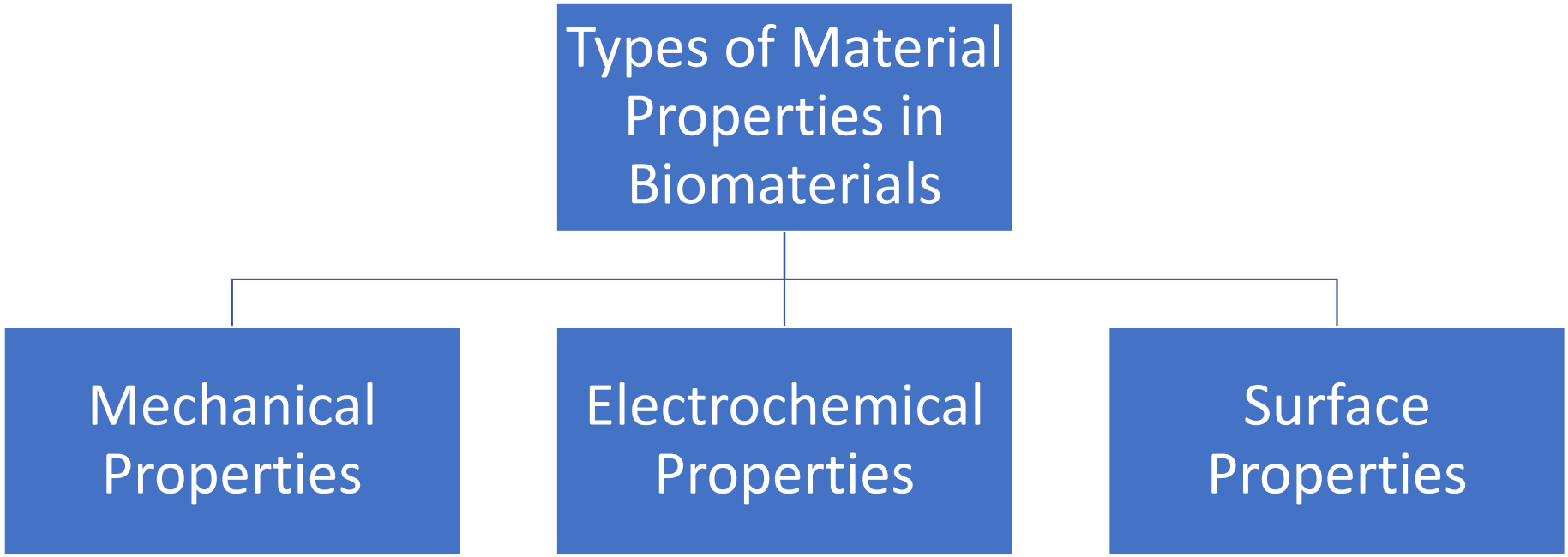


- These are qualitative descriptions (not precise)

Why Qualitative Is NOT Enough

- Qualitative terms are not reliable for:
  - Safety-critical systems
  - Medical devices
  - Structural design
- Example:
  - Engineers cannot build bridges using "strong material" only
- *Instead → use scientific measurements*

Types of Material Properties in Biomaterials



1) Mechanical Properties (Bulk Behavior)

- Describe how a material reacts to forces
- Include:
  - Deformation (change in shape)
  - Strength (resistance to force)
  - Fracture (breaking)
- Example:
  - Hip implants must tolerate loads up to several times body weight during walking
  - So, materials must be strong and durable

2) Electrochemical Properties

- Interaction with biological environment
- Describe corrosion behavior
- Important for metals inside the body
  - Example:
  - Stents in blood vessels → must resist corrosion in blood

3) Surface Properties

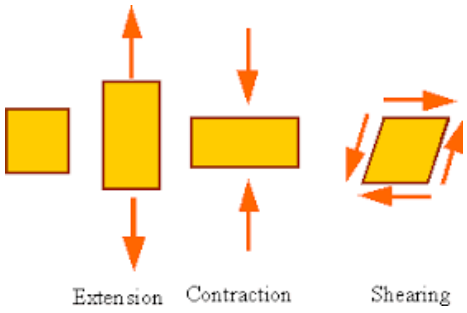
- Include:
  - Surface energy
  - Hydrophilicity / Hydrophobicity
- Control:
  - Cell attachment
  - Protein interaction
- Example:
  - Implant surfaces affect how cells stick or reject the material

Mechanical properties

Mechanical properties

- The response of a material to deforming forces is characterized by its mechanical properties.

- These forces can be of various types such as tensile, compressive, shear / torsional or combinations of these forces



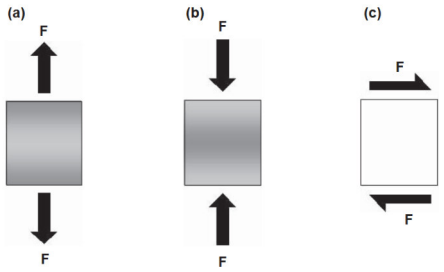
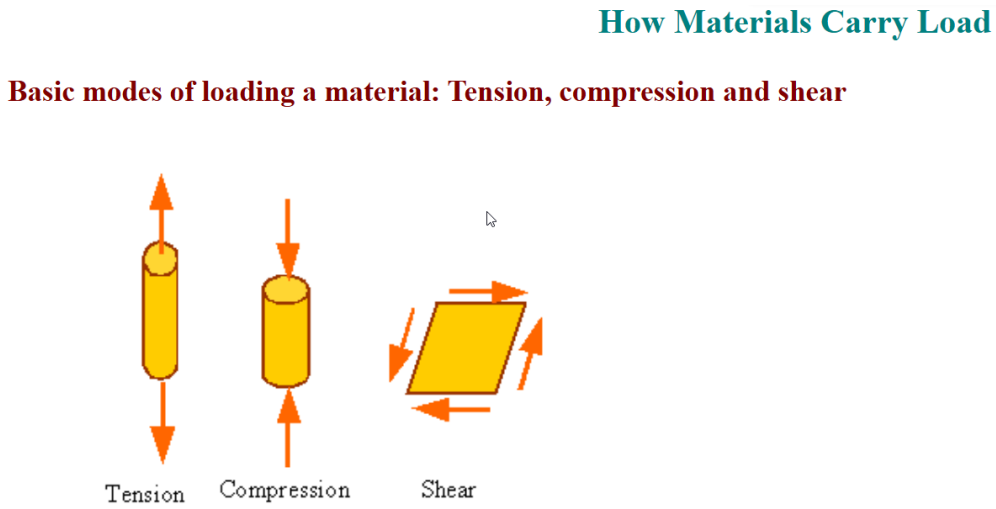


Figure 2.1
Different types of loading: (a) tensile; (b) compressive; (c) shear.

Very good summary→

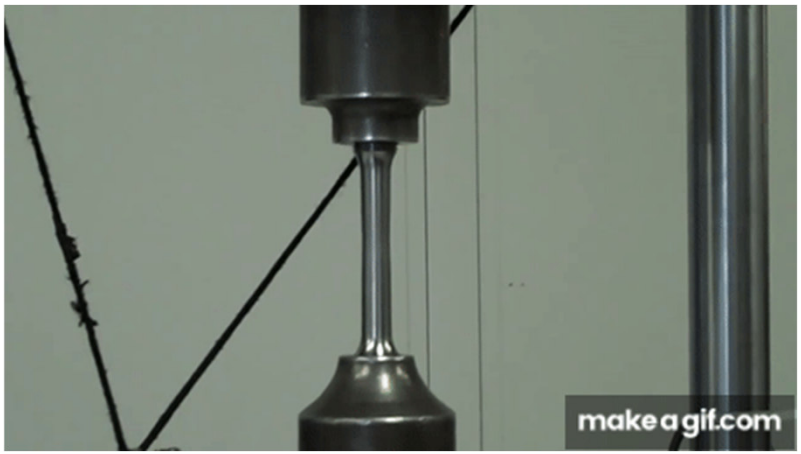
- https://emweb.unl.edu/NEGAHBAN/Em325/01-How-Materials-carry-load/How%20Materials%20Carry%20Load.htm

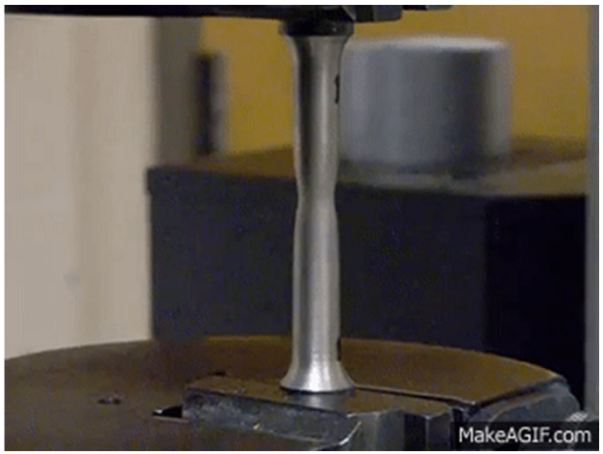


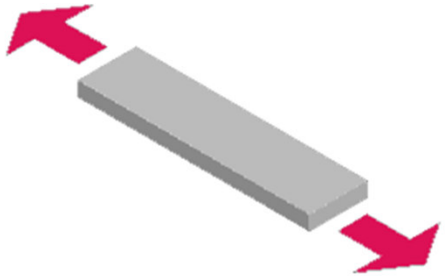
Mechanical properties

- Mechanical properties describe how materials deform under force
- Forces may be tensile, compressive, torsional, or combined
- Response depends on force type, loading rate, temperature, and environment
- Measured using force - deformation tests
- Tests are usually uniaxial, but can also be multiaxial or dynamic
- Some tests are time-dependent
- Standards like ASTM and ISO ensure reliable comparison

Tensile Test

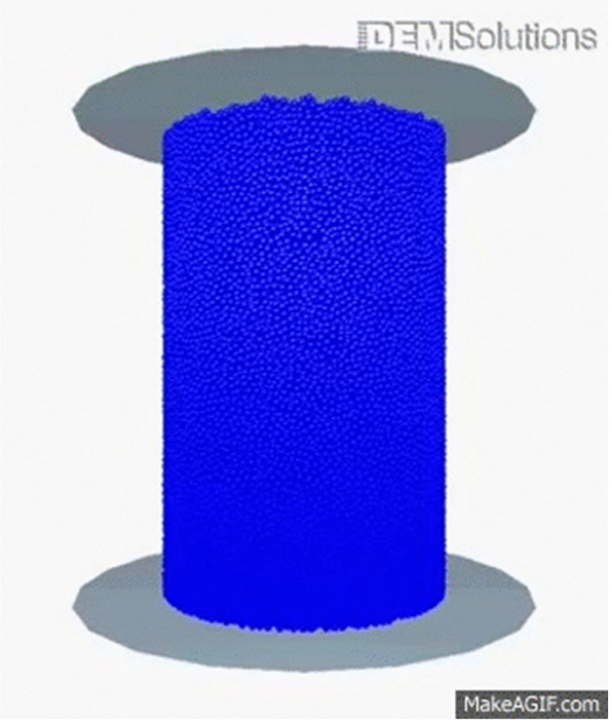






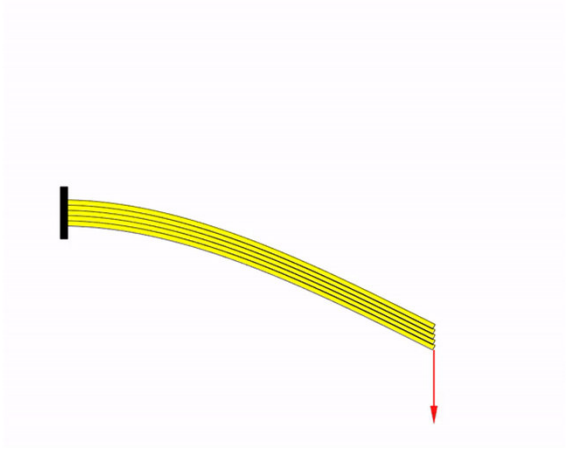


Compressive Test

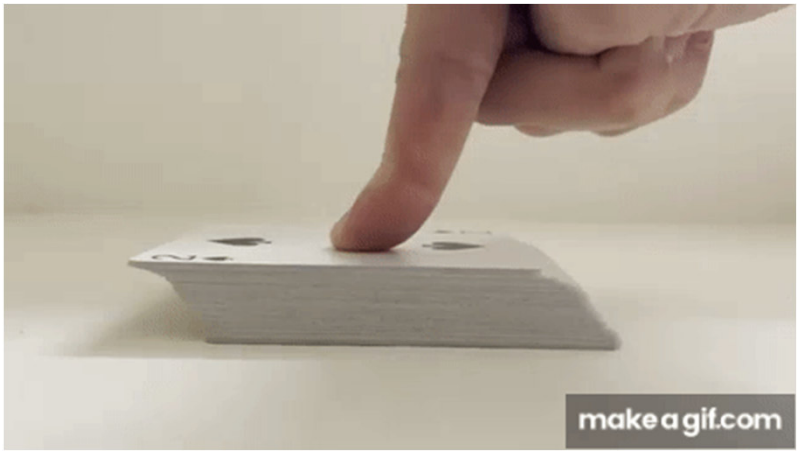


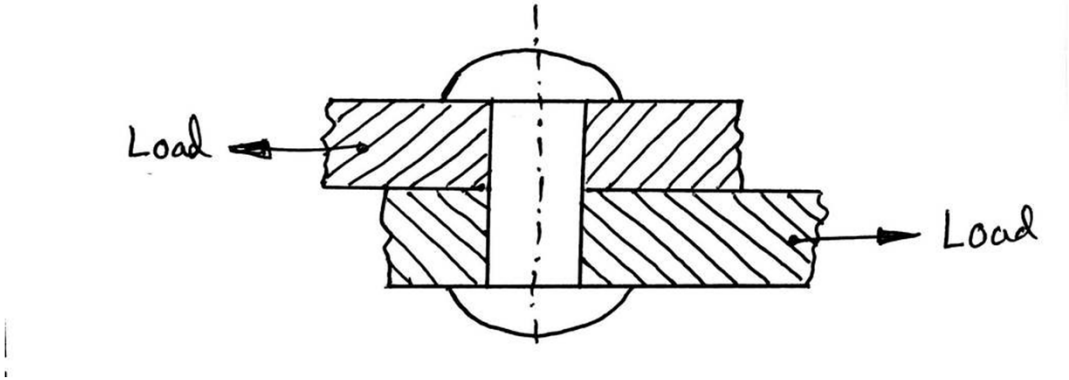


Sheer



ExtruDesign.com





Tensile testing : Basic test setup

- In a tensile test, the specimen is clamped at both ends between two jaws.
- The lower jaw is fixed, while the upper jaw moves upward.
- As shown in Figure 2.2, the upper jaw is connected to the machine crosshead through a load cell, which measures the applied force.
- The specimen may have:
  - Straight-sided bar shape
  - Dog-bone shape for rectangular sections
  - Dumbbell shape for circular sections
- These specimen shapes are shown in Figure 2.3.

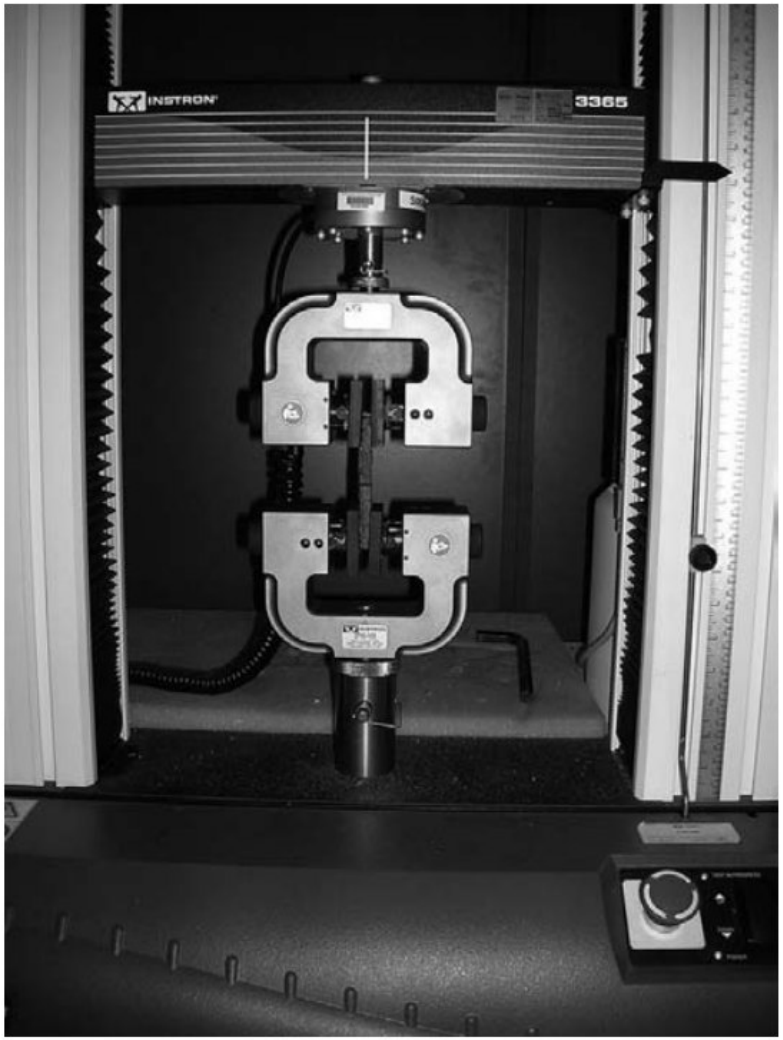


Figure 2.2
A tensile testing machine.

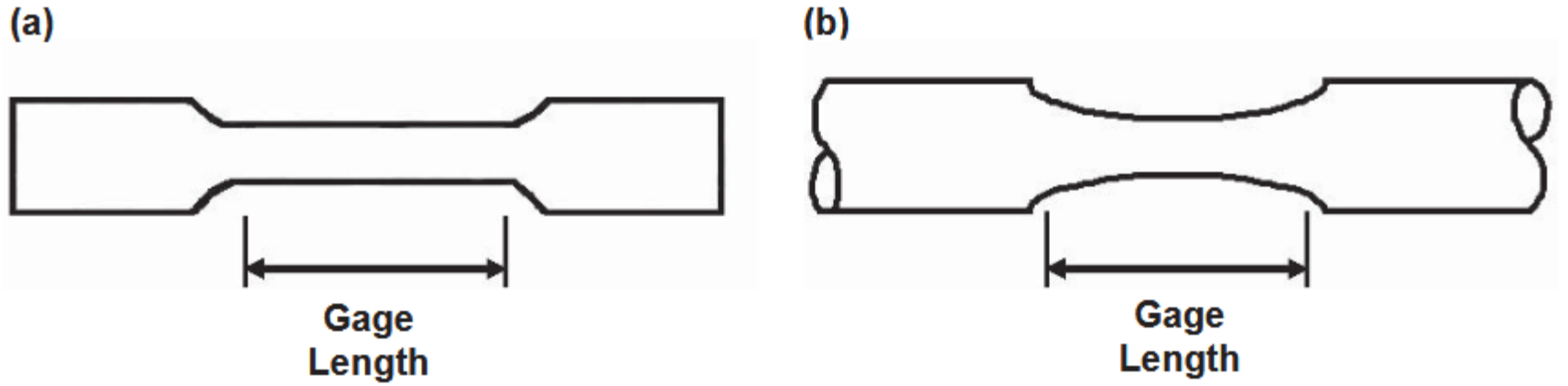
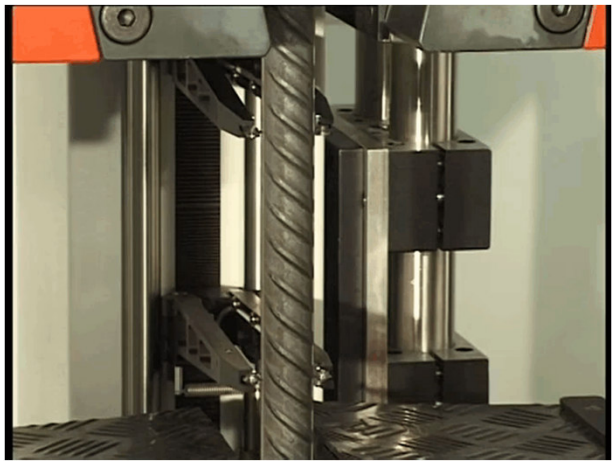


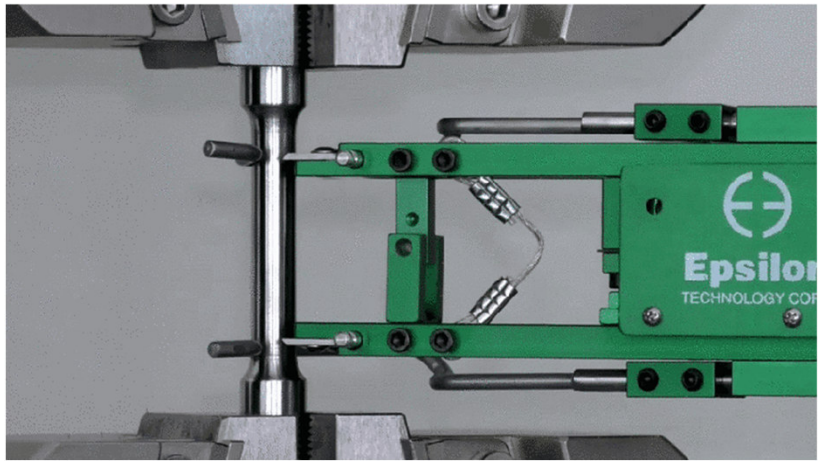
Figure 2.3

Standard types of specimens for testing the mechanical properties of a material using a tensile testing machine: (a) dog bone and (b) dumbbell.

Tensile testing : Measuring deformation

- An extensometer is used to measure the change in specimen length along the loading direction.
- The test gives data for force versus deformation.
- This data is then converted into a stress-strain curve.





Tensile testing : Stress and strain

- Stress means the applied force divided by the original cross-sectional area.
  - It tells us how concentrated the force is.
  - Unit: force per unit area
- Strain means the change in length divided by the original length.
  - It tells us how much the material deforms relative to its original size.
  - It is dimensionless (has no unit)
- These are defined in Eqs. (2.1) and (2.2).

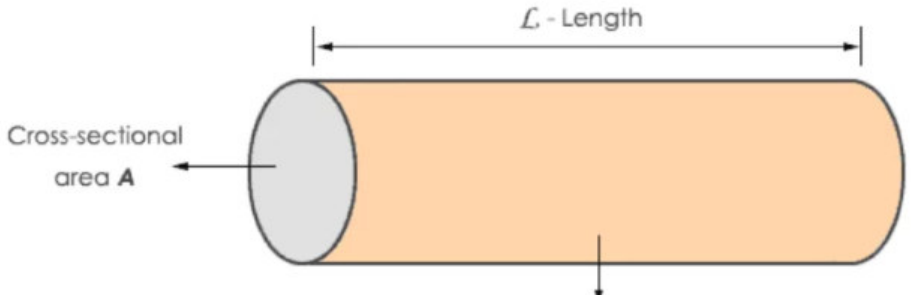
Stress – Strain

$$\text{stress } (\sigma) = \frac{\text{force}}{\text{original cross-sectional area}}, \tag{2.1}$$

$$\text{strain } (\varepsilon) = \frac{\text{change in length}}{\text{original length}}. \tag{2.2}$$

Tensile testing :Gage section

- The gage section is the part of the sample used for measurement.
- It is used to determine:
  - Cross-sectional area
  - Original length
- As shown in Figure 2.4:
  - For straight specimens, the gage section is the region between the jaws.
  - For dog-bone or dumbbell specimens, it is the reduced middle section.



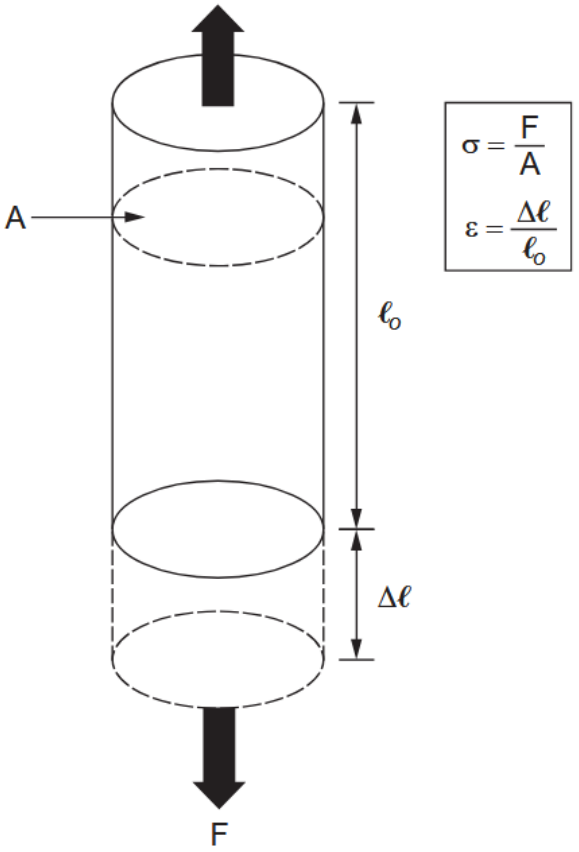


Figure 2.4

A rod under tensile loading showing stress and strain.

Tensile testing : Force is not the same as stress

- Force and stress are different.
- The same force can create different stress values depending on the area over which it acts.
- Example:
  - A knife pressed with its flat side causes low stress
  - The same force applied through the sharp edge causes very high stress and cuts the apple

Tensile testing : Stress–Strain Curve - Hooke's law and elastic region

- Robert Hooke showed that for many solids, deformation is proportional to load in the initial region.
- This is called Hooke's law.
- It appears as the linear part of the stress–strain curve.
- An example stress–strain curve is shown in Figure 2.5.
- In this region, the material behaves elastically, meaning it returns to its original shape after unloading.

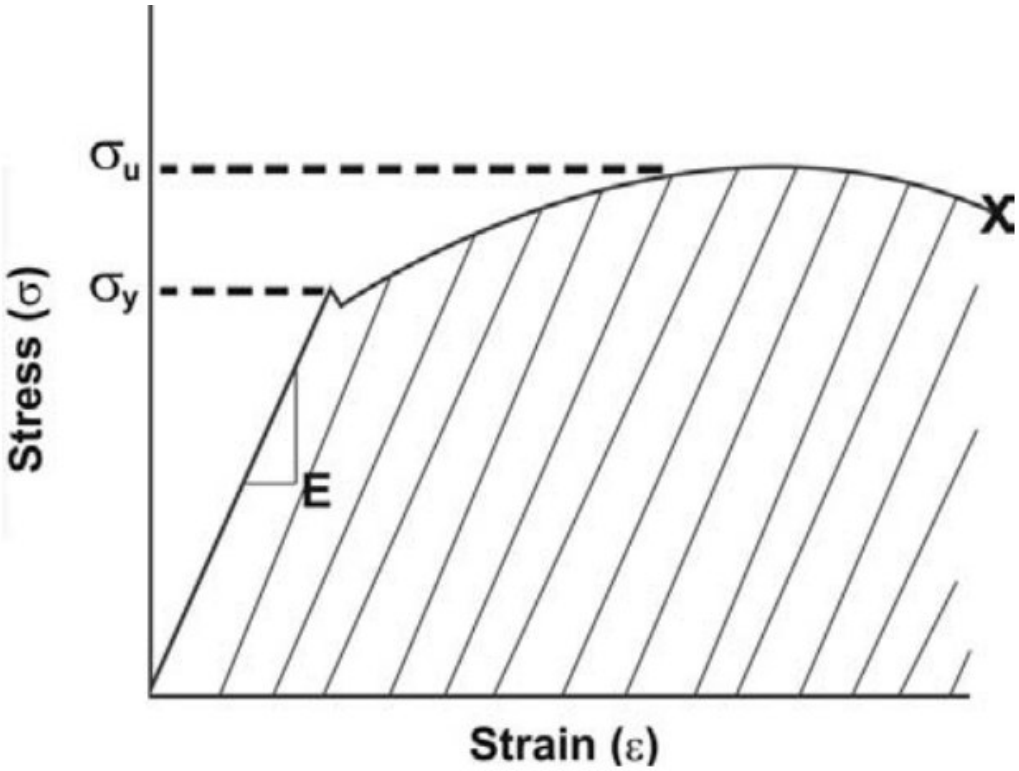


Figure 2.5

A stress–strain curve showing the yield stress (σ_y), the ultimate strength (σ_u), and the elastic modulus (E) given by slope of the linear portion. The shaded area under the curve represents the toughness of the material.

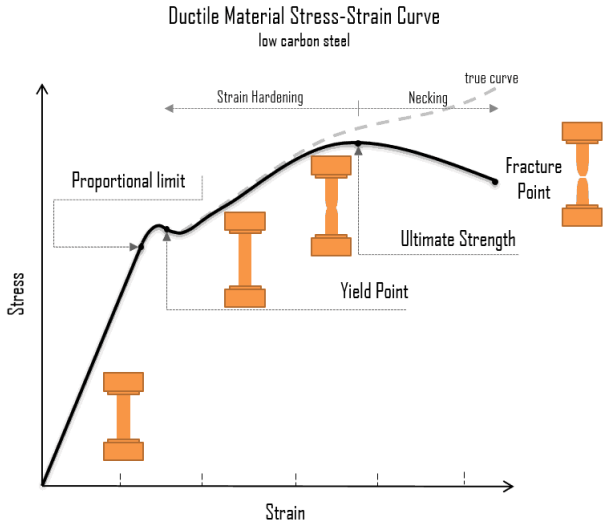
Tensile testing : Elastic modulus (E)

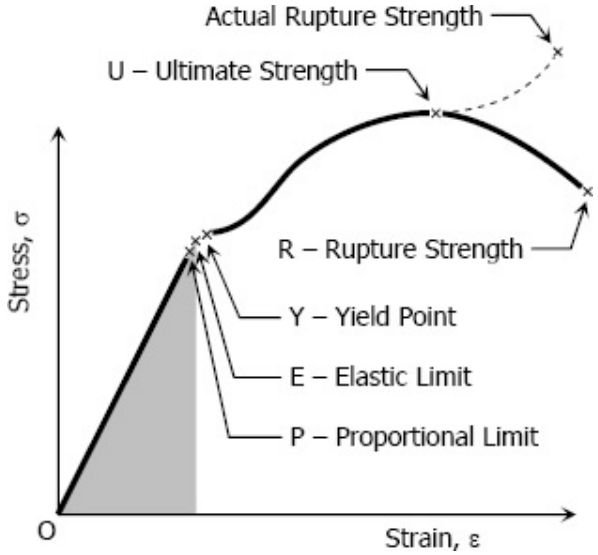
- The slope of the linear region is called the **elastic modulus** or Young's modulus.
- It measures the stiffness of a material.
- A **higher modulus** means the material is **stiffer** and **resists deformation more**.
- The relation is given in Eq. (2.3): **stress = modulus × strain**

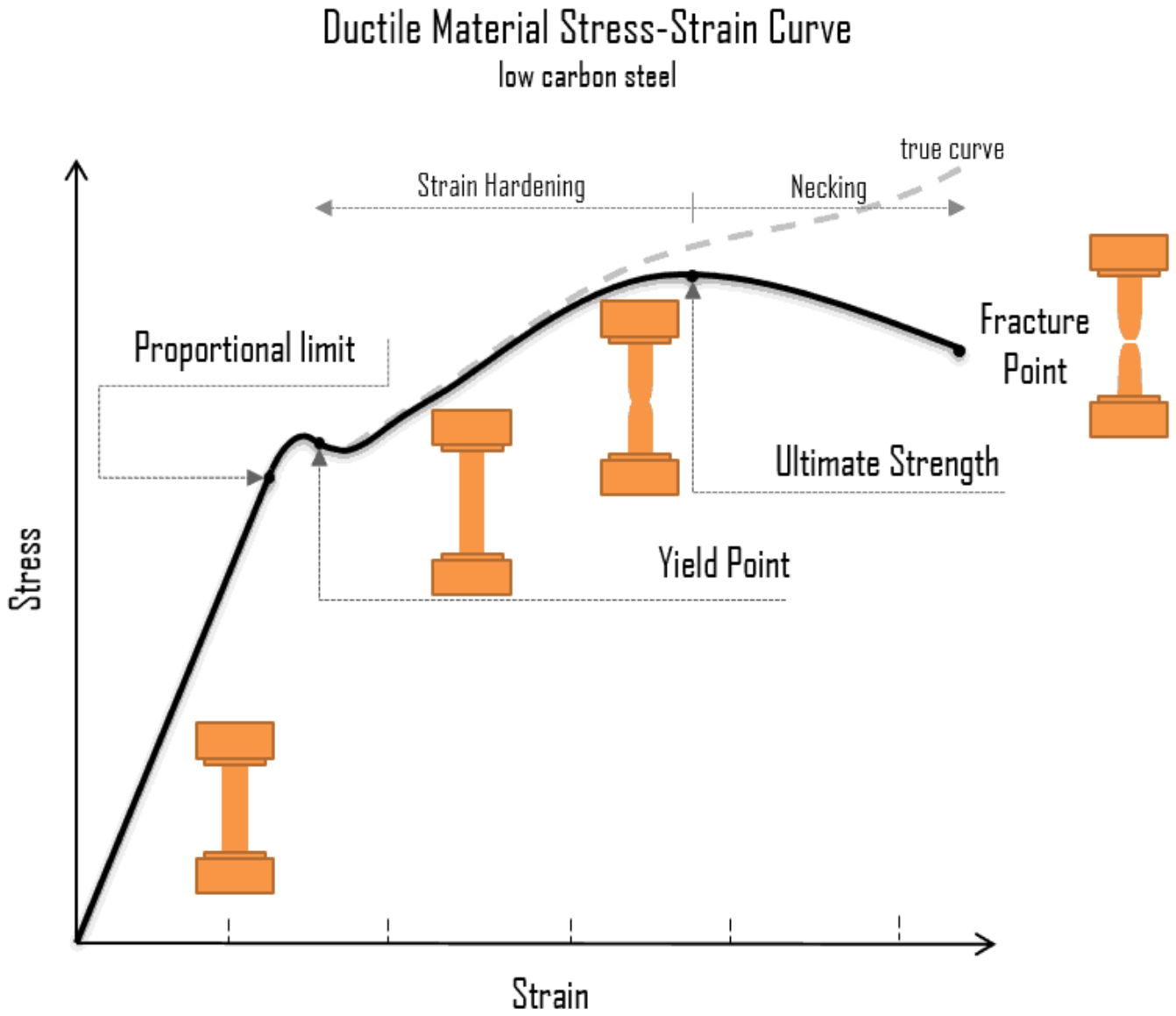
$$\sigma = E\varepsilon. \qquad (2.3)$$

Tensile testing :Proportional limit

- The proportional limit is the highest stress where stress remains linearly proportional to strain.
- Up to this point, the stress–strain curve stays straight.





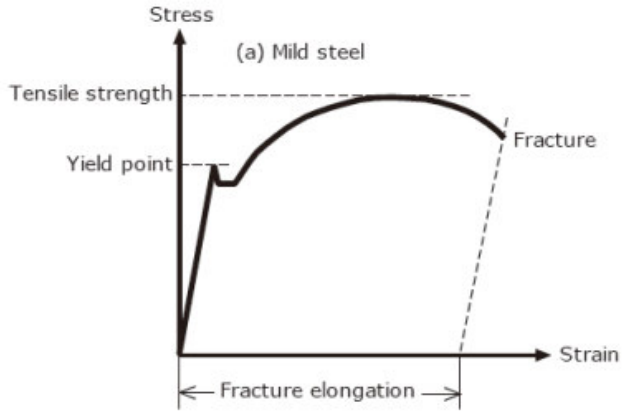

Ductile Material Stress-Strain Curve
low carbon steel
true curve
Strain Hardening
Necking
Proportional limit
Fracture
Point
Ultimate Strength
Yield Point
Stress
Strain

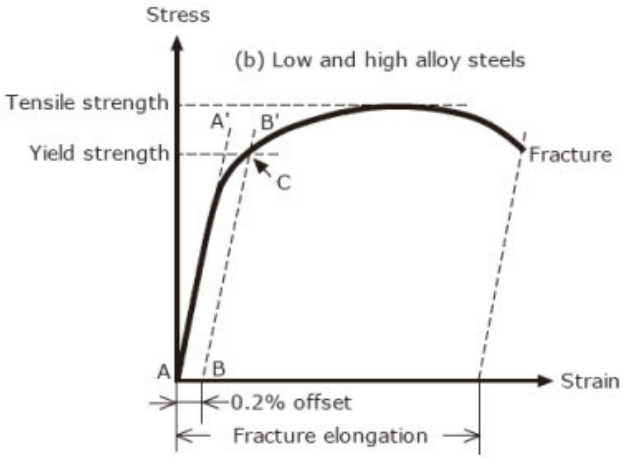
Tensile testing :Yield stress / yield point

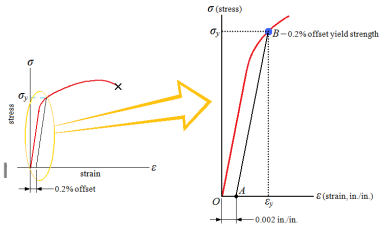
- The yield stress is the stress at which permanent deformation begins.
- This point is called the yield point.
- It may also be called:
  - Yield strength
  - Elastic limit
- After this point, the material does not fully return to its original shape after unloading.

Tensile testing :0.2% offset method

- Some materials do not show a clear yield point.
- In that case, yield stress is estimated using the 0.2% strain method.
- A line parallel to the linear elastic region is drawn starting at 0.2% strain.
- The intersection of this line with the stress-strain curve gives the yield point.

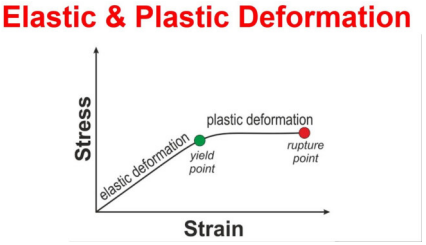






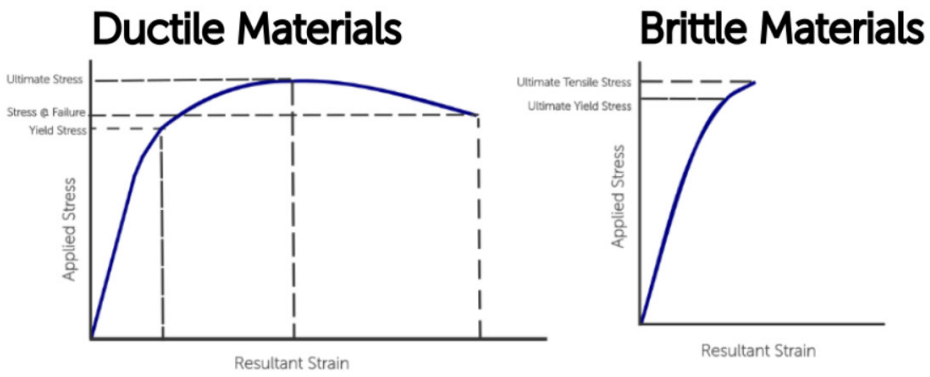
Tensile testing : Elastic and Plastic Behavior -
Elastic deformation and Plastic deformation

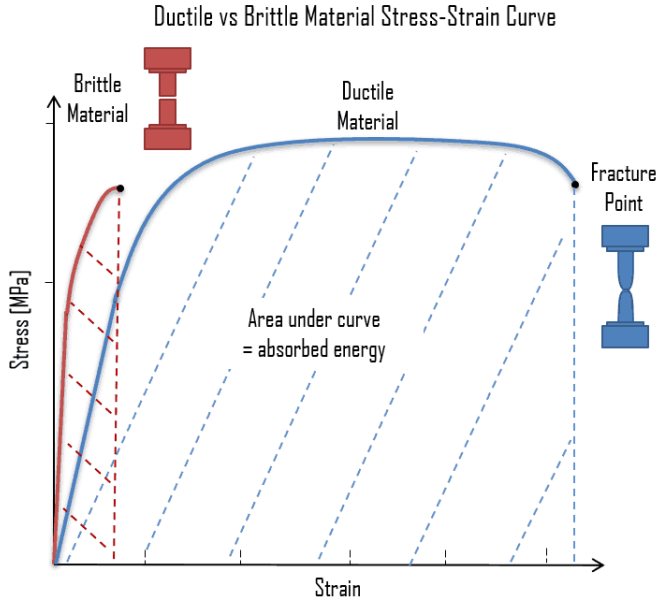
- Elastic deformation
  - In the elastic region, deformation is reversible.
  - When the load is removed, the material returns to its original dimensions.
- Plastic deformation
  - Beyond the elastic limit, the material enters plastic deformation.
  - Plastic deformation is permanent.
- It happens because of movement of:
  - Atoms/dislocations in metals
  - Molecular chains in polymers



Tensile testing : Brittle materials

- Brittle materials such as ceramics usually:
  - Do not show significant plastic deformation
  - Do not show a clear yield point
  - Fracture suddenly





Tensile testing : Important properties from the stress-strain curve - UTS and Toughness

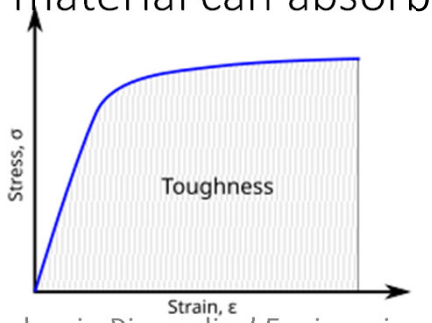
- Ultimate tensile strength (UTS)
  - The ultimate tensile strength is the maximum stress reached before fracture.
  - It represents the highest stress the material can withstand in tension.
- Toughness
  - Toughness is the ability of a material to absorb energy before fracture.
  - It is measured by the area under the stress-strain curve up to failure.
  - A tougher material can absorb more energy before breaking.

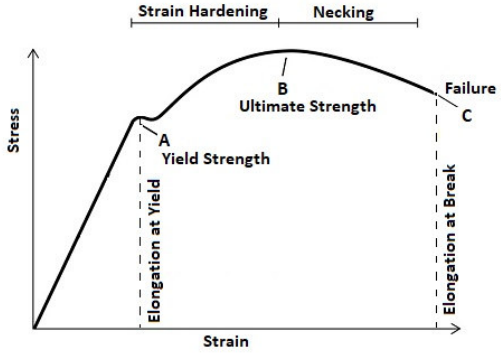
The modulus of toughness is given by:

$$U_t = \int_0^{\epsilon_f} \sigma \, d\epsilon$$

Where

- U_t is the modulus of toughness,
- σ is the stress,
- ϵ is the strain, and
- ϵ_f is the strain at fracture.



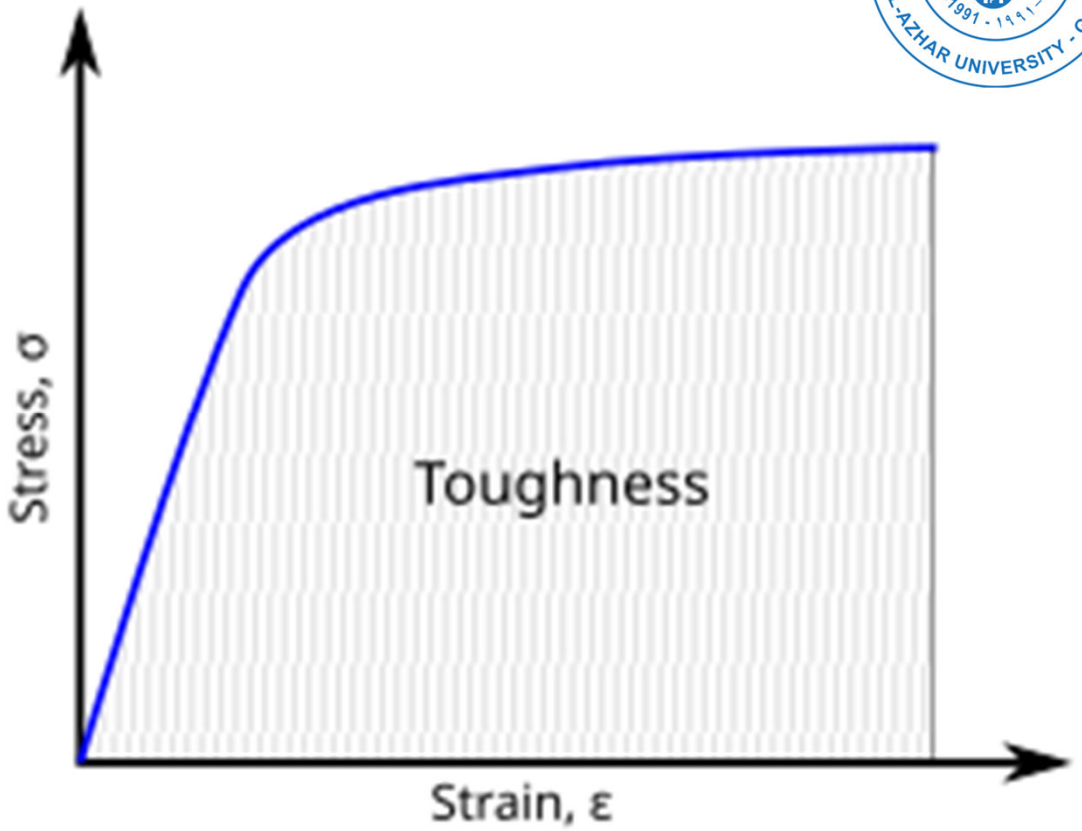


The modulus of toughness is given by:

$$U_t = \int_0^{\epsilon_f} \sigma \, d\epsilon$$

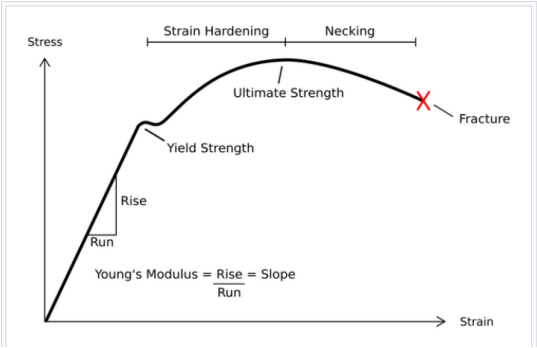
Where

- U_t is the modulus of toughness,
- σ is the stress,
- ϵ is the strain, and
- ϵ_f is the strain at fracture.



Tensile testing : Strain hardening - Strain hardening (work hardening)

- In many ductile metals, stress continues to rise after yielding.
- This is called **strain hardening**.
- It means the material becomes more resistant to further deformation.
- It happens because dislocations interact and reduce the mobility of atomic planes.
- Stress increases until the material reaches ultimate strength, then fracture follows.



Tensile testing : Engineering stress vs true stress

- Engineering stress:
- Engineering stress is calculated using the original cross-sectional area:
  - σ = F / A(original)
- This is the stress usually used in design.
- True stress:
- True stress uses the instantaneous actual area during deformation.
- Since the specimen area changes while stretching, true stress gives a more realistic value during loading.

Gfood reference

- https://yasincapar.com/engineering-stress-strain-vs-true-stress-strain/#:~:text=True%20stress%20is%20the%20applied,cross%2Dsectional%20area%20of%20material.

$\sigma = F/A_0$	Engineering Stress
$\sigma_t = F/A$	True Stress
$\varepsilon = \delta/L_0$	Engineering Strain
$\varepsilon_t = \ln(L/L_0)$	True Strain

F	Load
A_0	Cross-sectional area of specimen before deformation has taken place
A	Cross-sectional area of specimen at which the load is applied
δ	Total elongation
L_0	Original value of the gage length
L	Successive values of the length as it changes

Tensile testing : Behavior of different material classes - Polymers

- Usually have:
  - Low elastic modulus
  - Large deformation before fracture
  - Short or unclear linear region
  - Relatively low ultimate strength
  - High toughness

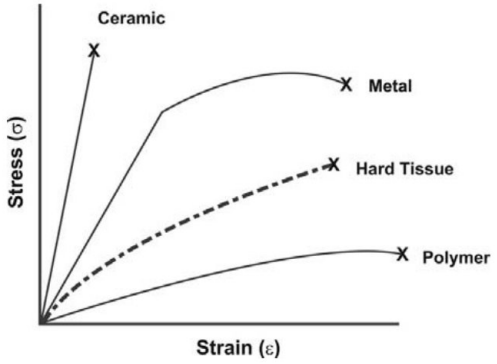


Figure 2.6
Representative stress–strain curves for ceramics, metals, hard tissue, and polymers.

- In some polymers, slope may increase during loading because molecular chains align.
- Their general behavior is shown in Figure 2.6.

Tensile testing : Behavior of different material classes - Metals

- Usually have:
  - Higher elastic modulus
  - Clear linear region
  - Well-defined yield point
  - Plastic deformation after yield
  - High ultimate strength

Refer to Figure Q1, differentiate the ceramic, polymer and metallic materials in terms of their toughness, strength and ductility.

(6 marks, C3)

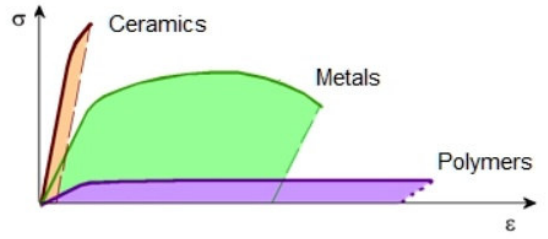


Figure Q1: Stress-strain curve

Tensile testing : Behavior of different material classes - Ceramics

- Usually have:
  - Very high elastic modulus
  - Very small deformation
  - High strength
  - Very low toughness
  - Brittle failure

Refer to Figure Q1, differentiate the ceramic, polymer and metallic materials in terms of their toughness, strength and ductility.

(6 marks, C3)

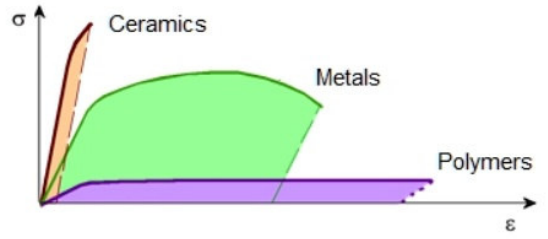
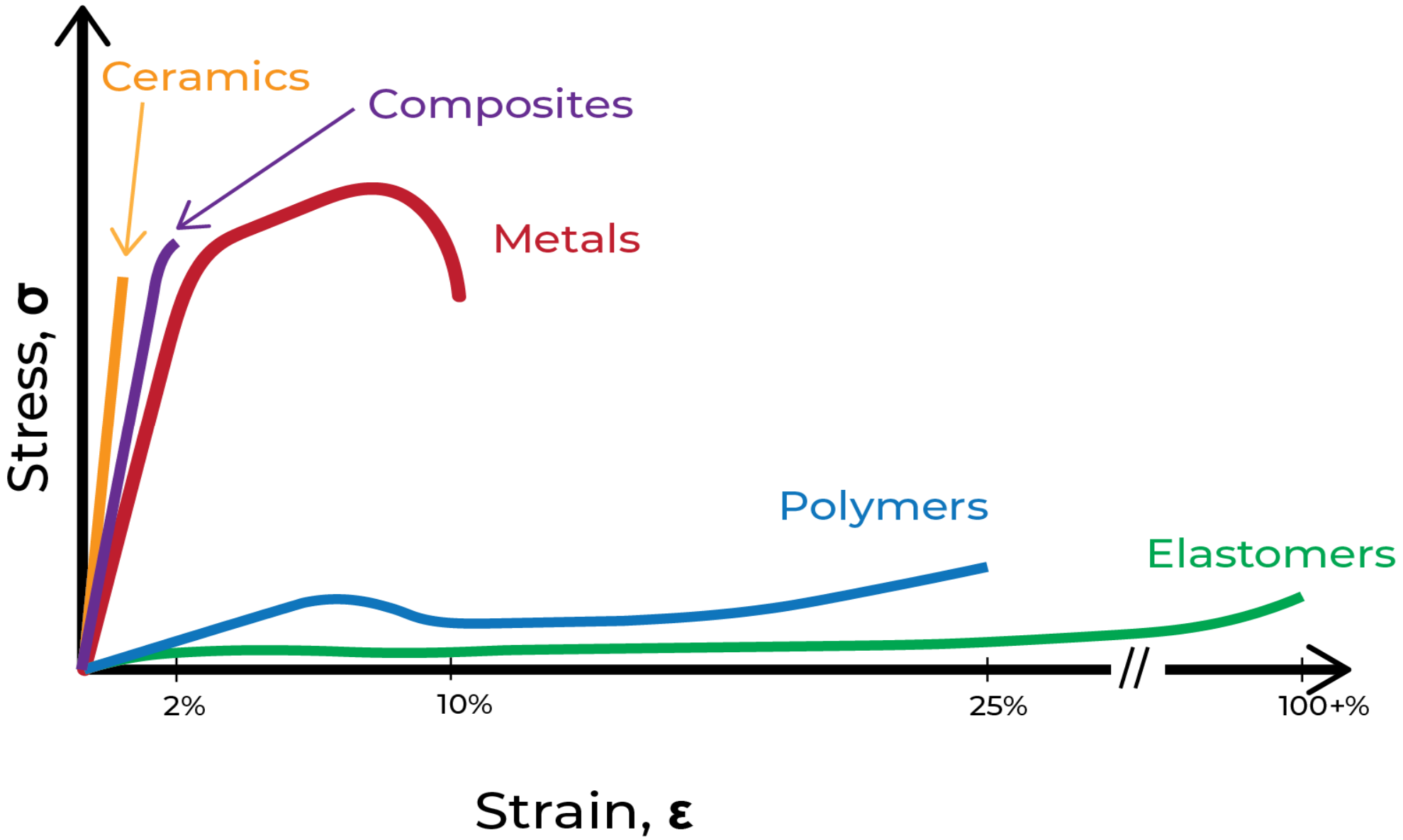


Figure Q1: Stress-strain curve


Ceramics
Composites
Metals
Polymers
Elastomers
Stress, σ
2%
10%
25%
100+%
Strain, ε

Refer to Figure Q1, differentiate the ceramic, polymer and metallic materials in terms of their toughness, strength and ductility.

(6 marks, C3)

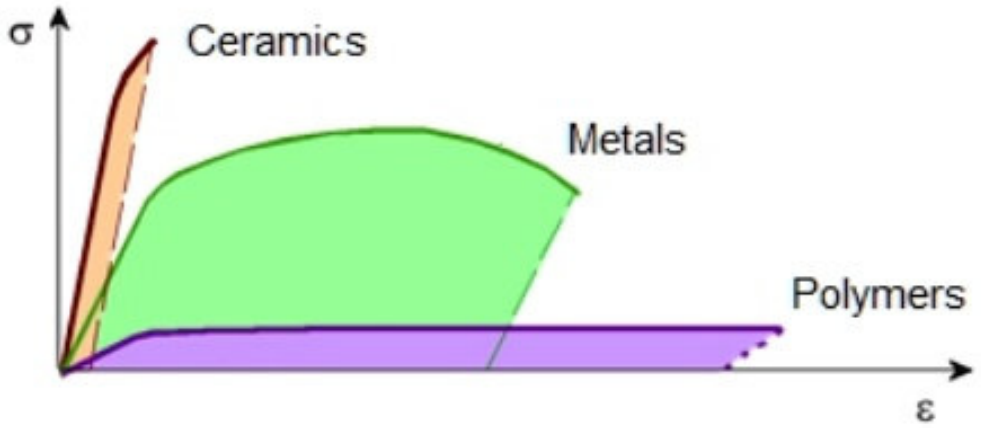


Figure Q1: Stress-strain curve

Answer ?

- Ceramics
- **High strength** because they resist deformation under load.
- **Very low ductility** because they fracture before much stretching happens.
- **Low toughness** because toughness depends on absorbing energy, and ceramics break suddenly with little strain.

Answer ?

- Metals
- **High strength** because they can withstand high stress.
- **High ductility** because they can deform a lot before breaking.
- **Highest toughness** because they combine both high stress and large strain, so they absorb a lot of energy before fracture.

Answer ?

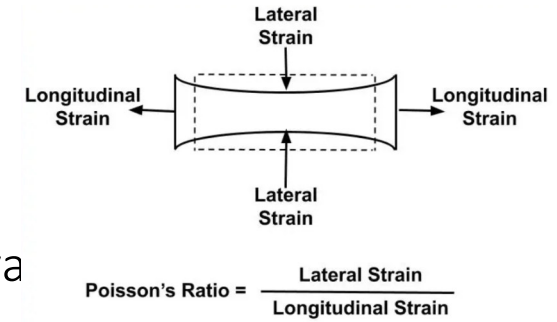
- Polymers
- Low strength because they cannot carry as much stress as metals or ceramics.
- High ductility because they can stretch a lot.
- Moderate toughness because they deform a lot, but the stress they carry is relatively low.

Answer ? – Summary

- Ceramics: strong but brittle, so they have low toughness.
- Metals: strong and ductile, so they have the highest toughness.
- Polymers: not very strong but can stretch a lot, so they are ductile with moderate toughness.

Tensile testing :Poisson effect and Poisson's ratio - Poisson effect and Poisson's ratio (v)

- Poisson effect : When a material is stretched in one direction, it usually shrinks in the perpendicular direction.
- Example:
  - A tensile-loaded cylinder becomes longer and thinner
- This is called the Poisson effect.
- Poisson's ratio (v) : is the ratio of:
  - Transverse strain / Longitudinal strain
- It is denoted by ν (nu).
- It describes how much the material contracts sidewa



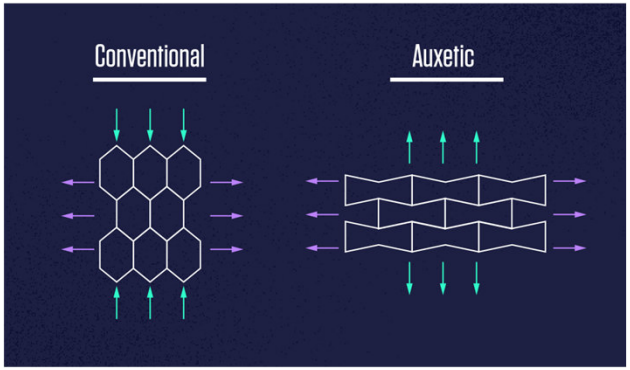
Tensile testing :Typical Poisson's ratio values

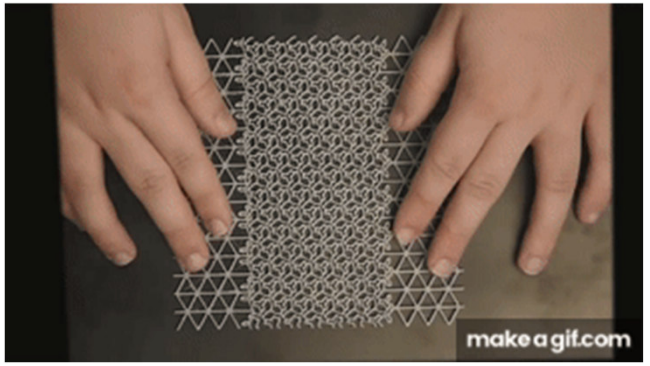
- For most isotropic simple materials: 0.2 to 0.5

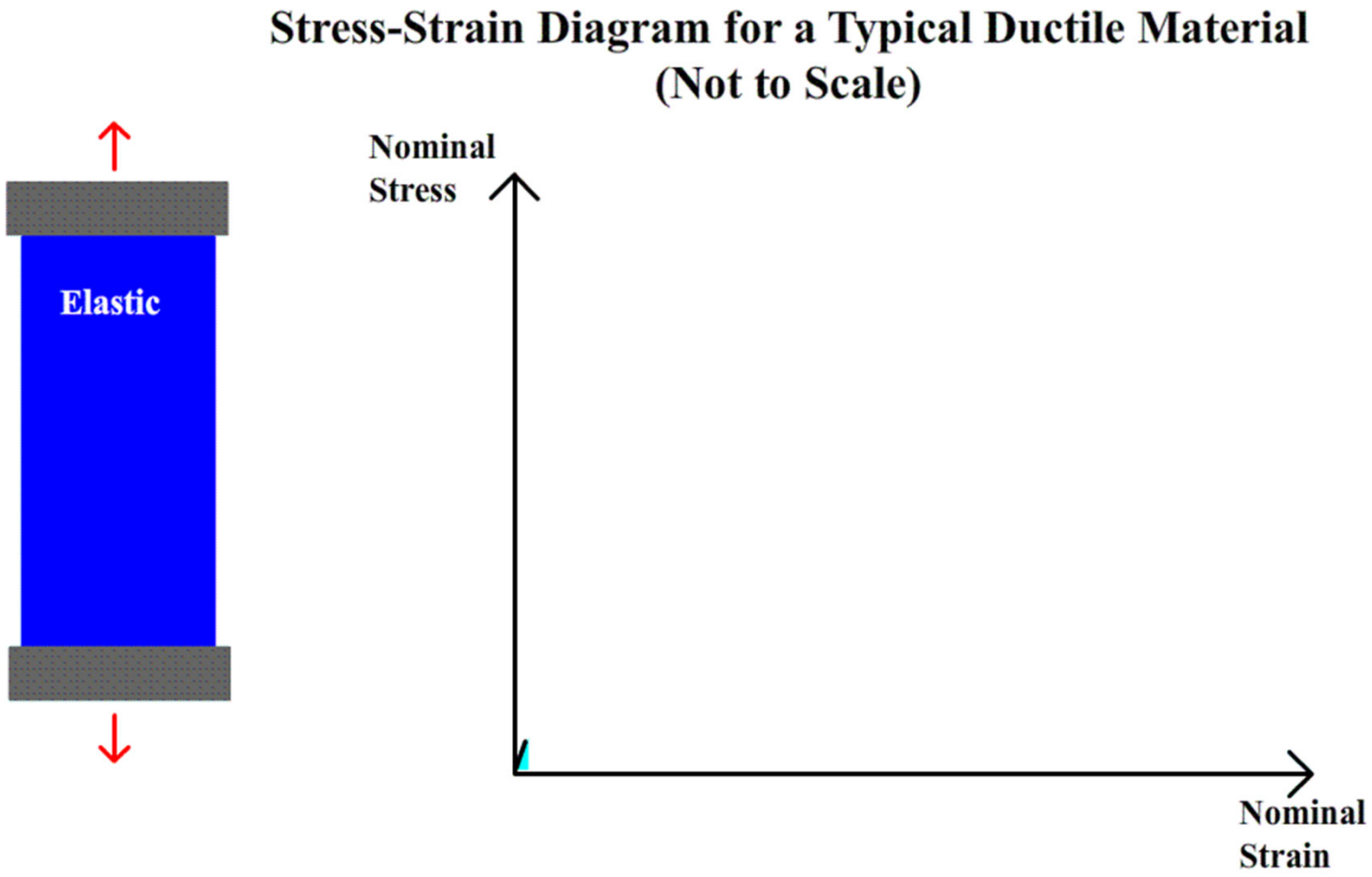
- Steel: about 0.3
- Rubber: nearly 0.5
- Cork: close to 0

Tensile testing :Auxetic materials

- Some special materials have negative Poisson's ratio.
- These materials become thicker sideways when stretched.
- They are called auxetic materials.
- This behavior is often seen in materials with hinge-like internal structures, such as some polymer foams.



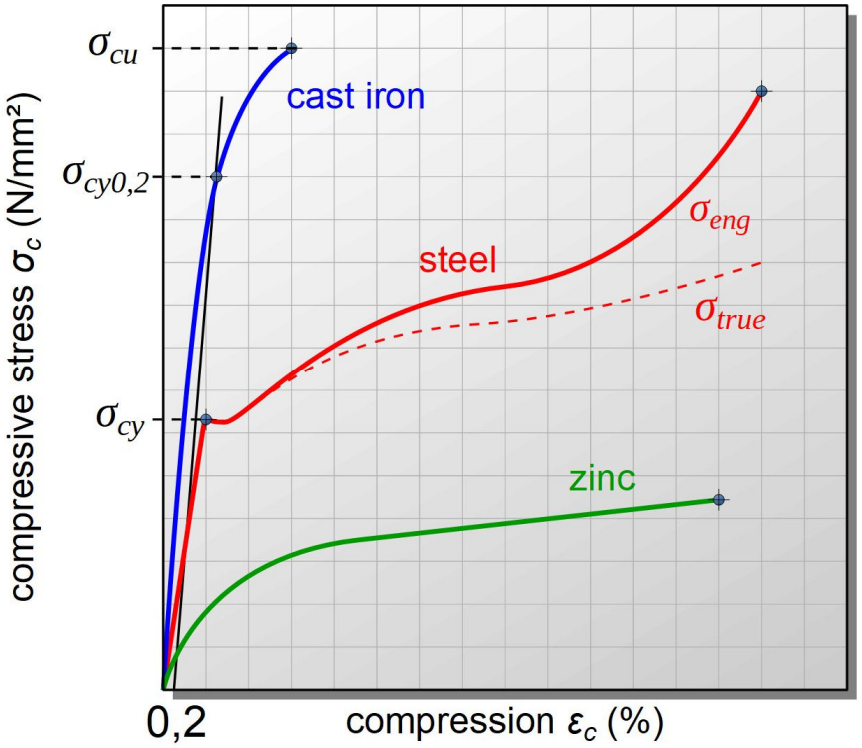



Stress-Strain Diagram for a Typical Ductile Material
(Not to Scale)
Elastic
Nominal
Stress
Nominal
Strain

Compressive Testing

- In compressive testing, block-shaped samples are used
- These are different from the specimen shapes used in tensile testing
- The same testing machine may be used, but with modified holders or fixtures
- The machine crosshead may move upward or downward to apply compressive load
- Material behavior in compression may differ from behavior in tension
- Elastic properties and failure modes can be different in compression and tension
  - Example: Ceramics are usually stronger in compression than in tension




σcu
σcy0,2
σcy
cast iron
steel
σeng
σtrue
zinc
compressive stress σc (N/mm²)
0,2
compression εc (%)

Shear Testing

- In shear testing, the load is applied parallel to the surface
- This is shown in Figure 2.7
- Deformation is measured in the same direction as the applied force

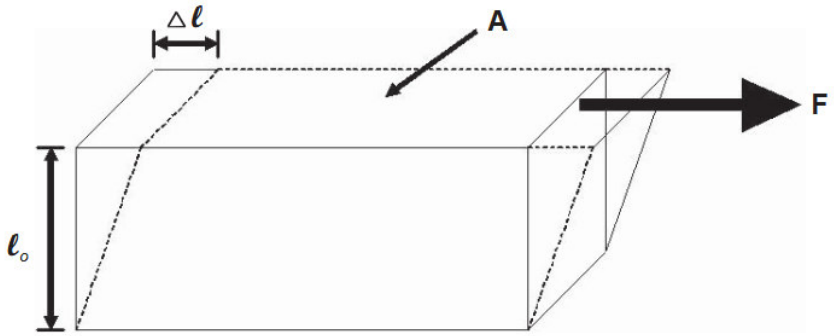
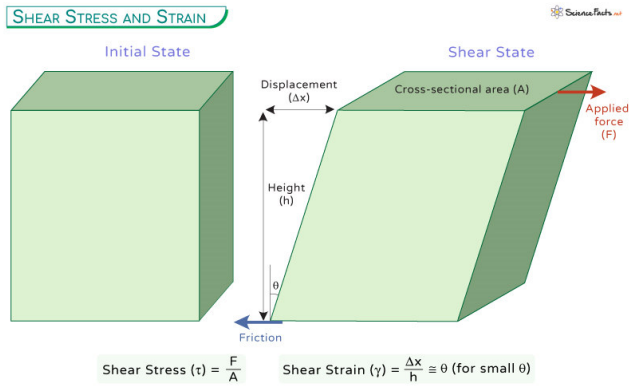


Figure 2.7
Deformation under shear loading.

Shear Testing : Main shear terms

- Shear stress is denoted by τ
- Shear strain is denoted by γ
- Shear modulus is denoted by G



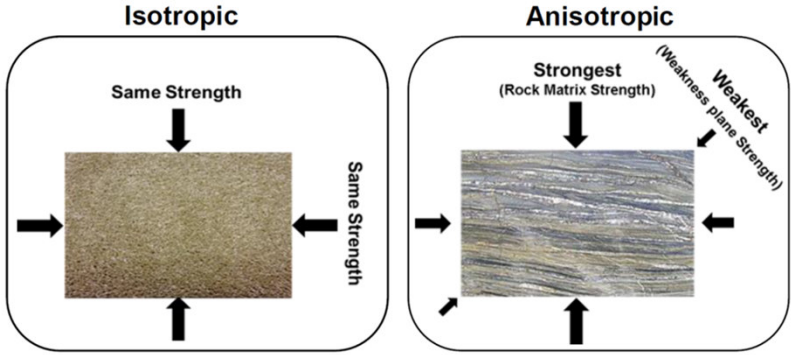
Shear Testing: Elastic relation

- In the elastic region, Hooke's law also applies to shear
- The relation between them is given in Eq. (2.4):
- $\tau = G\gamma$

$$\tau = G\gamma. \qquad (2.4)$$

Shear Testing : Isotropic vs anisotropic materials

- Isotropic materials have the same mechanical properties in all directions
- For isotropic materials, E and G are enough to describe stiffness
- Anisotropic materials have different properties in different directions
- In anisotropic materials, test results change with specimen orientation
  - Examples of anisotropic materials
  - Examples:
    - Bone
    - Ligaments
    - Skin
    - Cartilage



Bend or Flexural Tests:

- Samples can also be tested under bending load
- These samples have a uniform cross-section
- The cross-section may be:
  - Rectangular
  - Circular



Bend or Flexural Tests: Three-point bending test

- In a three-point test, the sample rests on two supports
- The supports are separated by a distance L
- A load is applied at the mid-point
- This causes the sample to bend or bow
- This setup is shown in Figure 2.8a

Bend or Flexural Tests:

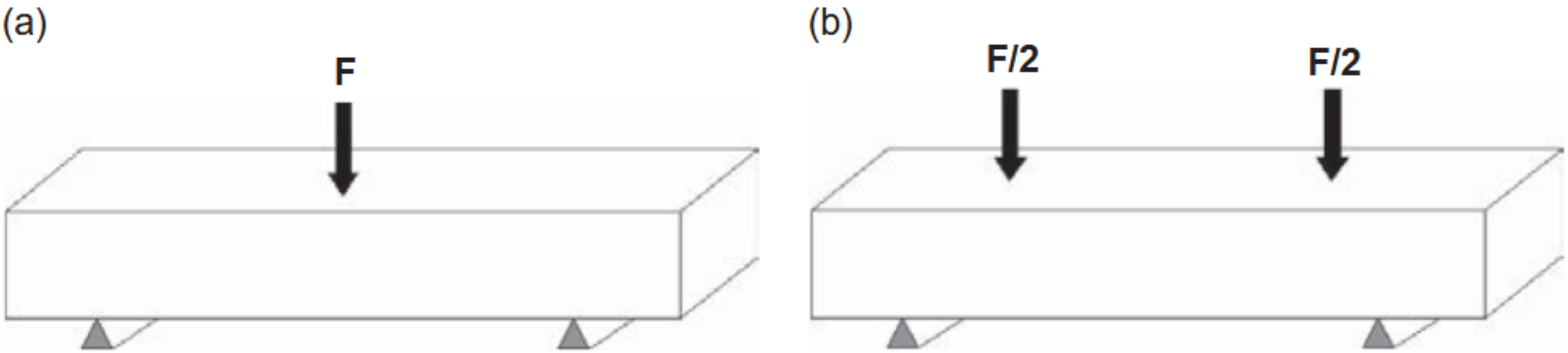
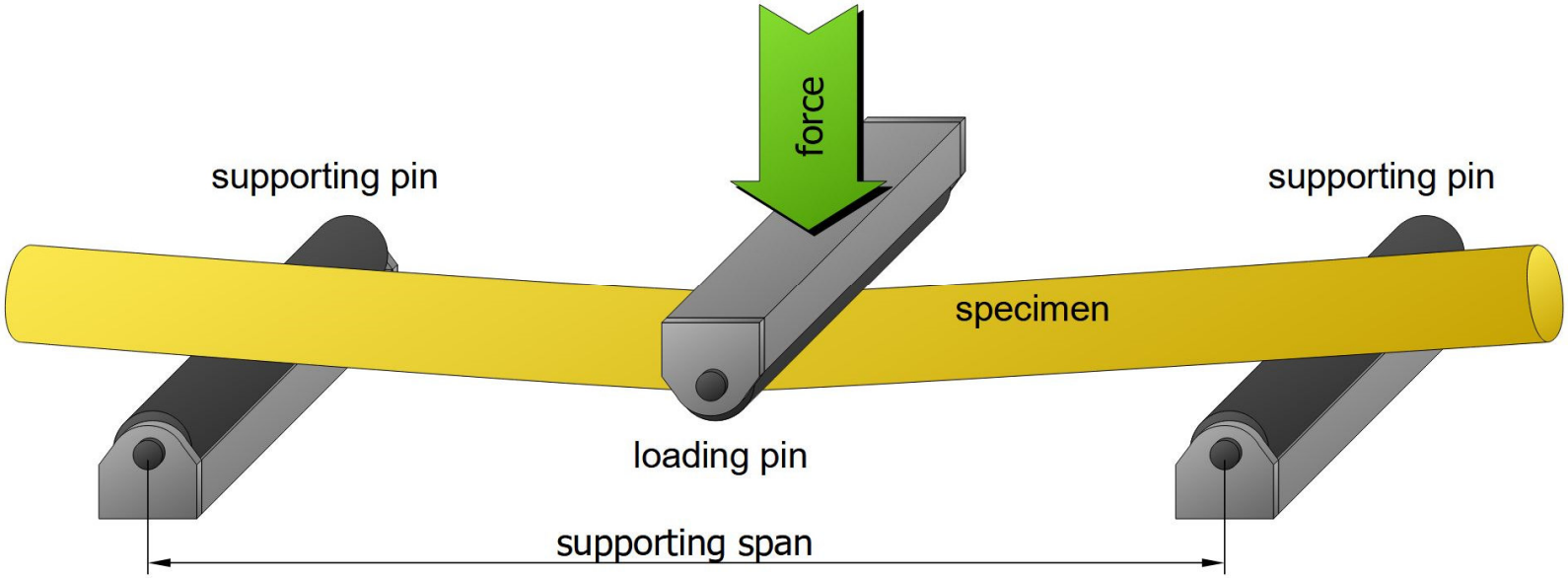


Figure 2.8

Bend tests: (a) three-point test, where the applied load is at one location; (b) four-point test, in which the applied load is at two places between the supports.

Bend or Flexural Tests: Stress during bending

- The inner side of the bent sample is under compression
- The outer side is under tension



Bend or Flexural Tests: Equations for rectangular beams

$$\sigma_f = 3FL/2bd^2, \quad (2.5)$$

$$\varepsilon_f = 6\Delta d/L^2, \quad (2.6)$$

where σ_f = stress in the outer material at the mid-point (MPa),
ε_f = strain of the outer material at the mid-point,
Δ = maximum deflection of the mid-point under the load (mm),
d = depth or thickness of the beam (mm),
b = width of the beam (mm),
L = length of the span between the two supports (mm), and
F = load applied at mid-point (N).

Bend or Flexural Tests: Four-point bending test

- In a four-point test, the load is applied through two points
- Each point applies F/2These loading points lie within the span L
- This setup is shown in Figure 2.8b

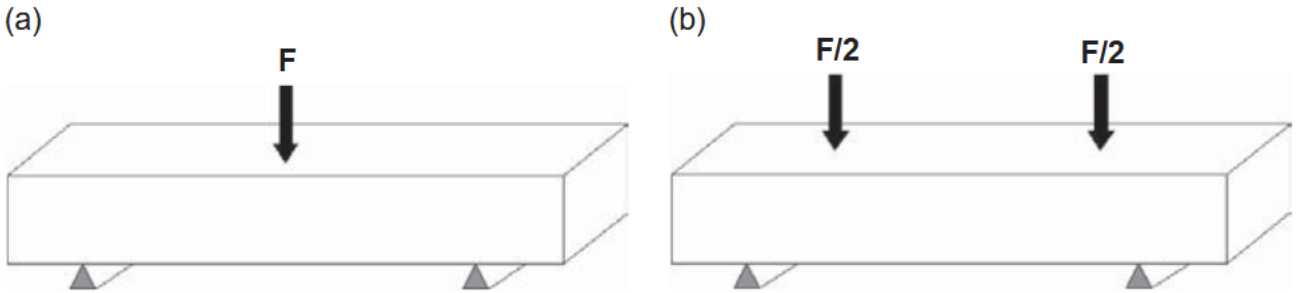


Figure 2.8

Bend tests: (a) three-point test, where the applied load is at one location; (b) four-point test, in which the applied load is at two places between the supports.

Bend or Flexural Tests: Main difference

- In the four-point test, the stress is constant between the two loading points
- The four-point test is better for determining material properties
- The three-point test is more useful for determining the overall strength of a sample or structure

Viscoelastic Behavior

- The previous properties assume deformation happens instantly when force is applied
- But some materials also show time-dependent deformation
- Viscoelastic behavior is the behavior of a material that shows both elastic recovery and time-dependent deformation.
- These materials are called viscoelastic materials

Viscoelastic Behavior : Viscoelastic model

- Viscoelastic materials can be represented by a combination of:
  - Spring → elastic behavior
  - Dashpot → viscous behavior
- This model is shown in Figure 2.9Figure 2.9a shows the dashpot model for viscous deformation
- Figure 2.9b shows the combined spring–dashpot model for elastic + viscous deformation

Viscoelastic Behavior

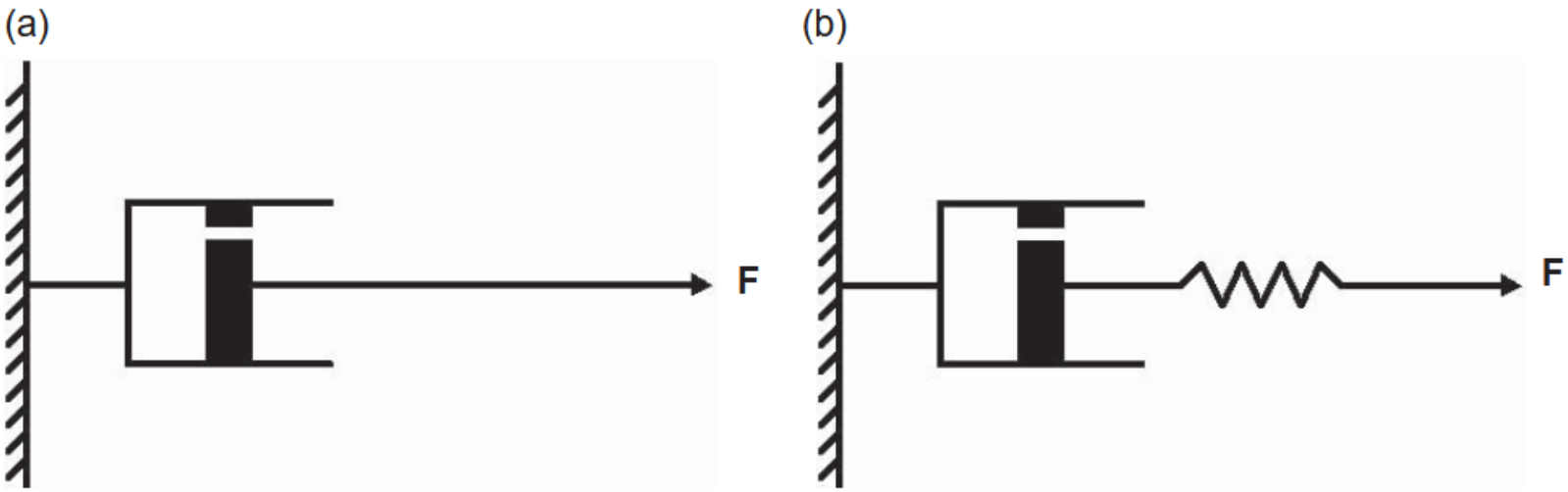


Figure 2.9

Models for a viscoelastic material: (a) cylinder and piston or dashpot model for viscous deformation; (b) dashpot and spring model for combined viscous and elastic deformation.

Viscoelastic Behavior: Main viscoelastic behaviors

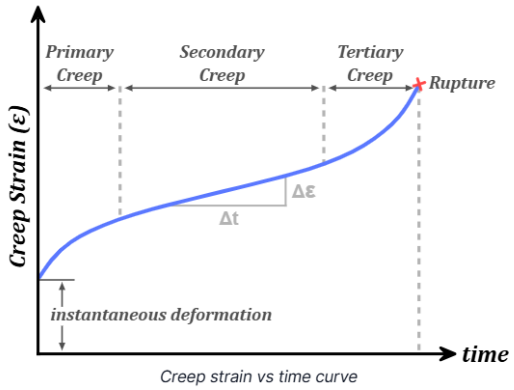
- Viscoelasticity appears mainly in:
  - Creep
  - Stress relaxation
- Viscoelastic behavior means the material shows both elastic and viscous response under load.
- In creep, the material continues to deform with time under a constant load.
- In stress relaxation, the stress decreases with time while the material is kept at a constant deformation.

Viscoelastic Behavior: Creep

- Creep occurs when a material is quickly loaded to a certain stress
- Then that stress is kept constant over time
- The material first shows:
  - Instantaneous elastic deformation
- Then it shows:
  - Additional time-dependent viscous deformation

Viscoelastic Behavior: Stages of creep

- Creep usually occurs in **three stages**, shown in **Figure 2.10**:
  - **Primary creep** → high deformation rate that slows quickly
  - **Secondary creep** → deformation becomes constant with time on a log scale
  - **Tertiary creep** → rapid deformation followed by failure



Creep strain vs time curve

Viscoelastic Behavior

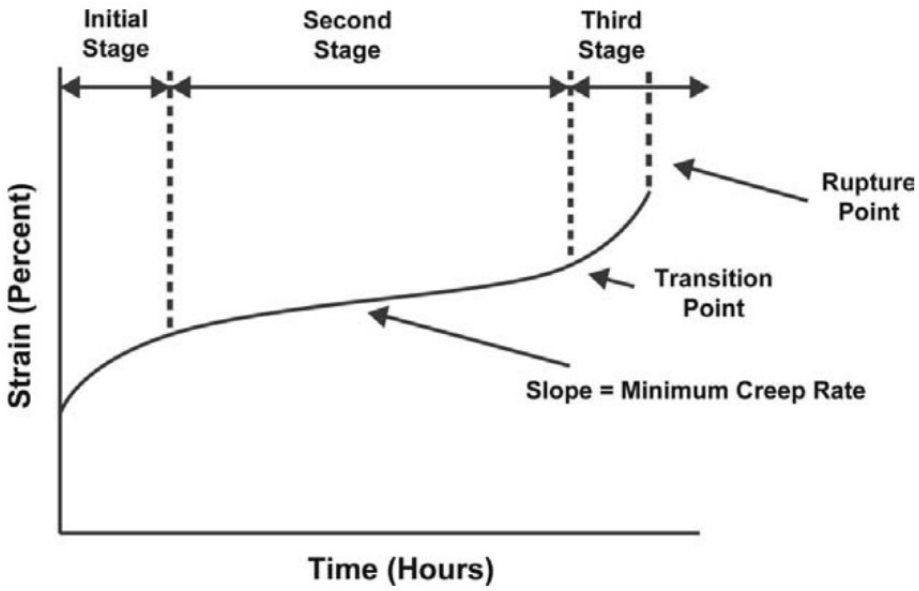


Figure 2.10

The different stages of creep deformation. Upon loading of a viscoelastic material there is initial elastic deformation followed by an increasing rate of strain in primary creep. This is followed by a linear creep strain–time relationship during secondary creep. Lastly, under tertiary creep, there is rapid deformation leading to failure.

Viscoelastic Behavior: Important notes about creep

- Not all three stages always appear
- High temperature is usually needed for tertiary creep
- Increasing temperature and stress increases creep
- Creep does not require a minimum stress level
- But it may not occur below a minimum temperature

Viscoelastic Behavior: Stress relaxation

- Stress relaxation occurs when a viscoelastic material is:
- Quickly deformed to a certain level
- Then held at constant deformation
- In this case, the stress decreases with time

Viscoelastic Behavior: Where viscoelasticity is seen

- Viscoelasticity is found in:
  - Most polymers
  - Some metals
  - Rarely in ceramics

Ductile and Brittle Fracture: What is failure?

- **Failure** can be defined in different ways depending on the application
- It may mean:
  - Onset of plastic deformation
  - Reaching a certain strain level
  - Final fracture
- **What is fracture?**
  - **Fracture** is the separation of a material into **two or more pieces**
  - It happens when the cohesive strength between atoms is exceeded by the applied stress

Ductile and Brittle Fracture: Main fracture types

- Fracture is mainly divided into two categories:
  - Ductile fracture
  - Brittle fracture
- Whether a material fails in a ductile or brittle way depends on:
  - Material properties
  - Temperature
  - Rate of loading

Ductile and Brittle Fracture: Ductile fracture

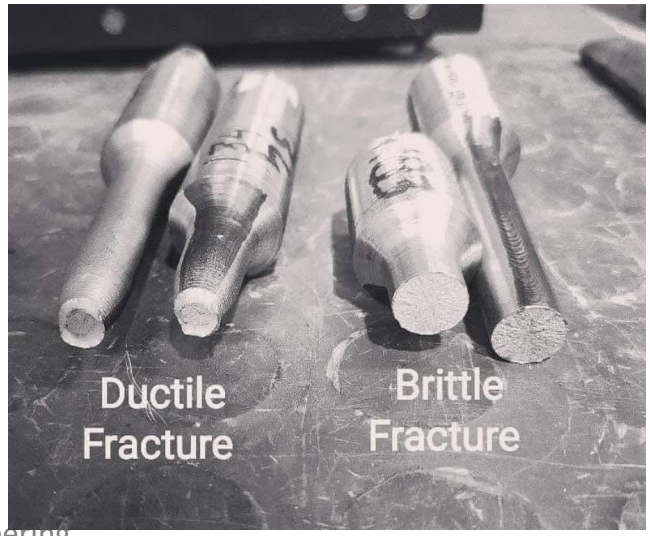
- Ductile fracture occurs after plastic deformation
- As stress increases, microdefects such as:
  - Voids
  - Cavities form inside the material or on its surface
- These defects then join together to form a crack
- The crack grows by absorbing more defects
- The crack usually travels perpendicular to the applied load

Ductile and Brittle Fracture: Final ductile failure

- When the crack reaches a critical size, it propagates rapidly
- In round specimens, fracture near the surface occurs at about 45° to the applied load
- This happens along the plane of maximum shear stress
- This produces the cup-and-cone fracture shape shown in Figure 2.11a

Ductile and Brittle Fracture: Microscopic appearance

- Under a scanning electron microscope, ductile fracture shows a dimpled surface
- These dimples represent the microdefects that merged to form the crack



Ductile and Brittle Fracture: Brittle fracture

- True brittle fracture occurs with no plastic deformation
- Failure happens by cleavage of atomic bonds along crystallographic planes
- The crack propagates perpendicular to the applied stress
- The fracture surface is usually macroscopically flat
- This is shown in Figure 2.11b

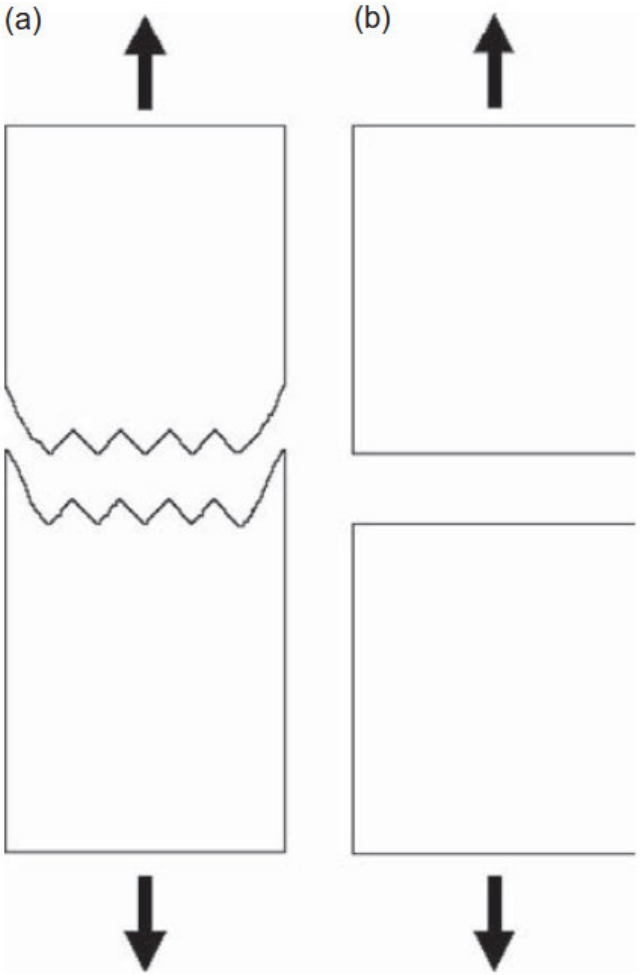


Figure 2.11

Types of fracture: (a) moderately ductile fracture showing a cup–cone failure surface; (b) brittle fracture usually presents a macroscopically flat surface.

Ductile and Brittle Fracture: Microscopic appearance

- Under the microscope, brittle fracture may show:
  - V-shaped chevron marks
  - Fan-like patterns spreading from the crack origin
- In very fine-grained materials, the surface may appear:
  - Shiny
  - Smooth
  - Without clear texture

Ductile and Brittle Fracture: Material examples

- Ceramics usually fail by brittle fracture
- Metals usually fail by ductile fracture
- But metals may become brittle after heat treatment
- Polymers may show either ductile or brittle fracture depending on whether they are:
  - Amorphous Glassy
  - Crystalline
- Even normally ductile materials can sometimes fail in a brittle manner
- This is explained by fracture mechanics

Stress Concentration: Theoretical vs actual failure

- Theoretically, a material should fail at about E/10
- But in real engineering tests, failure can occur at much lower stresses
- In some cases, failure happens at only E/10,000

Stress Concentration: Why this became important

- This difference became clear after major failures in the early 20th century
- Examples included:
  - Ships breaking into two under normal conditions
  - Aircraft losing wings in flight
- These failures happened even though the designs followed the engineering standards of that time

Stress Concentration: Griffith's idea

- In the 1920s, A. A. Griffith proposed that materials contain inherent flaws
- These flaws may be:
  - Microscopic voids
  - Small cracks
- Such flaws act as stress concentration points or stress risers

Stress Concentration: What stress concentration means

- Stress is not distributed uniformly near a flaw
- The applied stress becomes magnified at the flaw tip
- The sharper the flaw, the higher the stress concentration
- So:
  - Smaller tip radius = higher stress
  - Sharper crack = more dangerous

Stress Concentration: Defects do not have to be microscopic

- Stress concentration can also occur at larger discontinuities
- Examples:
  - A screw hole in a fracture fixation plate
  - A change in cross-section in the stem of a hip prosthesis

Stress Concentration: Why it causes failure

- The local stress at the defect may become much higher than the applied stress
- This local stress can exceed the material's cohesive strength
- So fracture may start at the defect even when the overall applied stress is still low

Stress Concentration: Equation

$$\sigma_t = k \cdot \sigma, \tag{2.7}$$

where σ = applied stress,

σ_t = stress at edge of defect, and

k = factor determined by shape and size of flaw or defect; k has a value of 2 for a hemispherical defect but can be very high for other cases depending on the sharpness (radius) of the crack tip.

Stress Concentration: During crack propagation

- A growing crack has a sharp tip
- This creates high stress ahead of the crack
- In ductile materials
  - The high stress causes plastic deformation at the crack tip
  - This blunts the crack
  - Blunting reduces the stress concentration
- In brittle materials
  - There is no plastic deformation
  - So the crack tip stays sharp
  - This keeps the local stress high and causes rapid failure

Stress Concentration: Compression and brittle materials

- Brittle materials are usually stronger in compression
- Under compression, crack tips are pressed together
- This reduces crack growth and slows propagation

Fracture Toughness: Stress near a flaw

- The stress near a crack can be described using the stress intensity factor, K
- K helps describe the stress at a point located:
  - At a distance r from the crack tip
  - At an angle θ from the crack tip

$$K = Y\sigma(a)^{1/2},$$

where Y = a constant based on the geometry of the specimen and the crack,
σ = the stress at (r, θ), and
a = the crack length.

Fracture Toughness: Critical value: fracture toughness

- The critical value of K for brittle fracture is called fracture toughness, Kc
- Brittle fracture occurs when K exceeds Kc

Fracture Toughness: Meaning of fracture toughness

- Fracture toughness is a material property
- It measures how well a material resists brittle fracture
- It is especially important when a flaw or crack already exists

Fracture Toughness: Importance in design

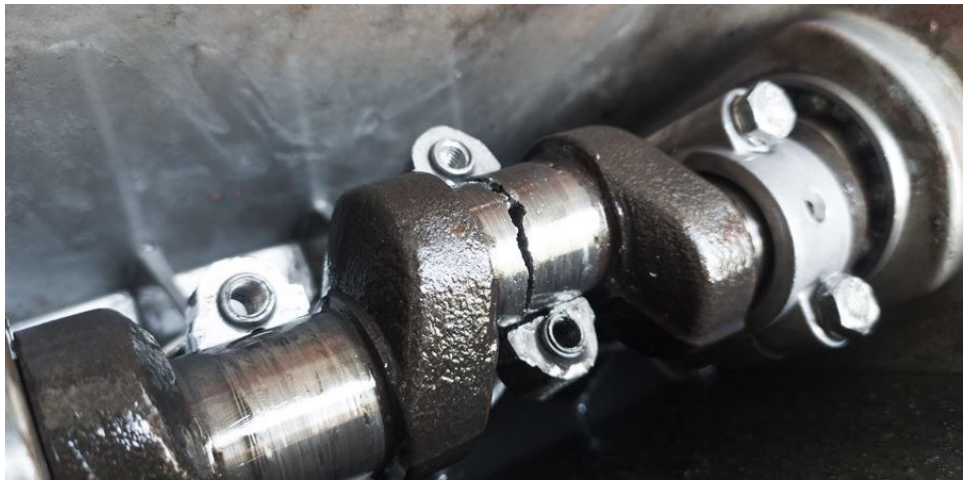
- In engineering design, it is not enough to consider only:
  - Ultimate strength
  - Yield strength
- It is also important to consider fracture toughness
- This helps prevent brittle failure

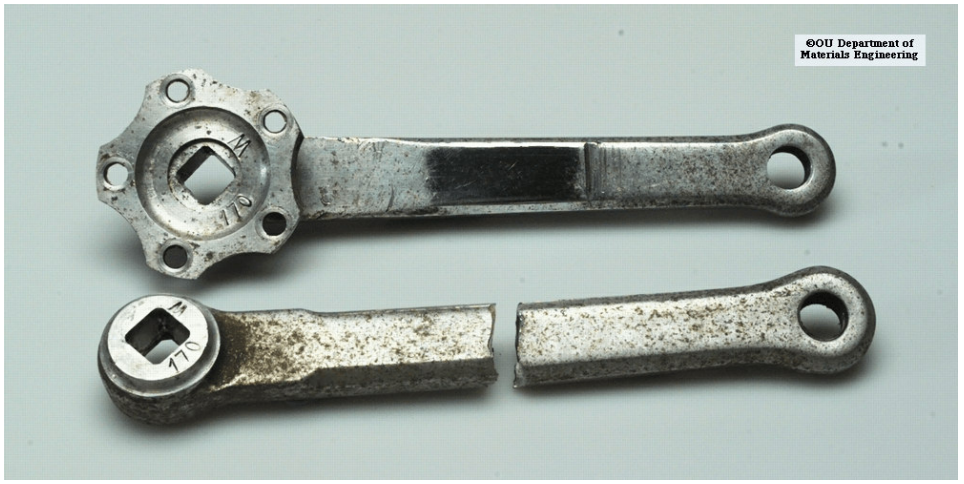
Fracture Toughness: Design consideration

- Fracture analysis may be based on:
  - The largest undetected defect that may exist in the material
  - A known notch, hole, or slit in the structure

Fatigue: What is fatigue?

- Under a single load, materials fail at their ultimate strength
- But under repeated loading cycles, they can fail at much lower stress
- This is called fatigue failure





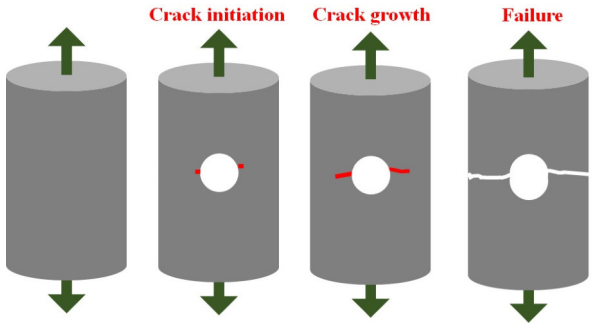
Fatigue: Loading in fatigue

- The load cycles do not need to fully unload the material
- The loading pattern may be:
  - Sinusoidal
  - Sawtooth
  - Uneven

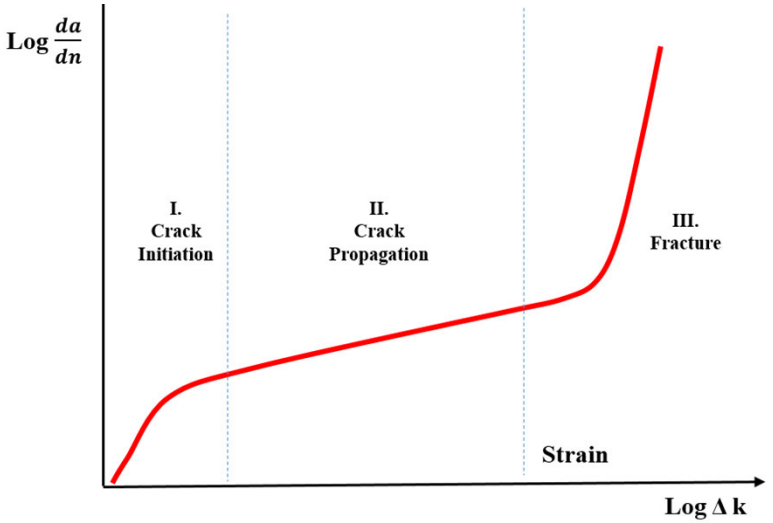
Fatigue: Why it matters in biomaterials

- Fatigue is very important for implants under cyclic loading
- Examples:
  - Total joint prostheses
  - Dental implants
  - Heart valves

Fatigue: Stages of fatigue failure



- 1. Crack initiation
  - Fatigue failure starts with **crack initiation**
  - Cracks usually start at:
    - Existing **stress concentration sites**
    - Imperfections caused by increasing **dislocations** in the crystal structure
- 2. Crack propagation
  - After initiation, the crack begins to **grow**
  - The crack tip acts as a **stress riser**
  - This creates high local stress at the tip
  - Crack growth occurs **intermittently**
  - The crack advances each time the material is loaded
- 3. Final fracture
  - Final fracture occurs when the crack reaches a **critical size**
  - At this stage, the **critical stress intensity factor** is reached



Fatigue: Strength, toughness, and fracture toughness

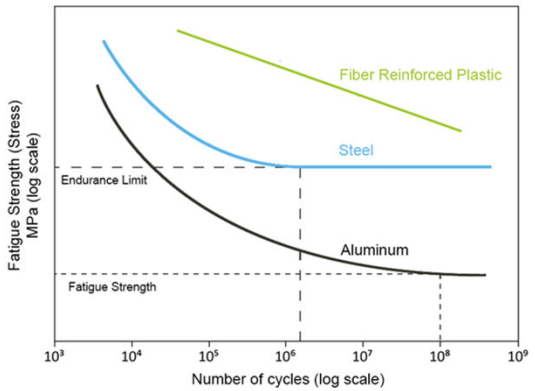
- Strength and fracture toughness are both material properties
- Failure occurs when:
  - Stress exceeds strength
  - Or stress intensity exceeds fracture toughness
- Important distinction
  - Fracture toughness is not the same as toughness
  - Toughness means the amount of energy required to cause failure

Fatigue: Fatigue fracture surface

- Materials that fail by fatigue show characteristic fracture patterns
- These include:
  - Striations
  - Wave-like patterns
- These patterns spread from the crack initiation point
- The spacing between striations indicates crack growth per loading cycle

Fatigue: S–N curve

- Fatigue behavior is shown using the stress–number of cycles to failure (S–N) curve
- This is shown in Figure 2.12
- In this graph:
  - S = stress
  - N = number of cycles to failure



- The number of cycles is plotted on a logarithmic scale
- Each point on the curve represents a 50% probability of failure at that stress-cycle combination

Fatigue:

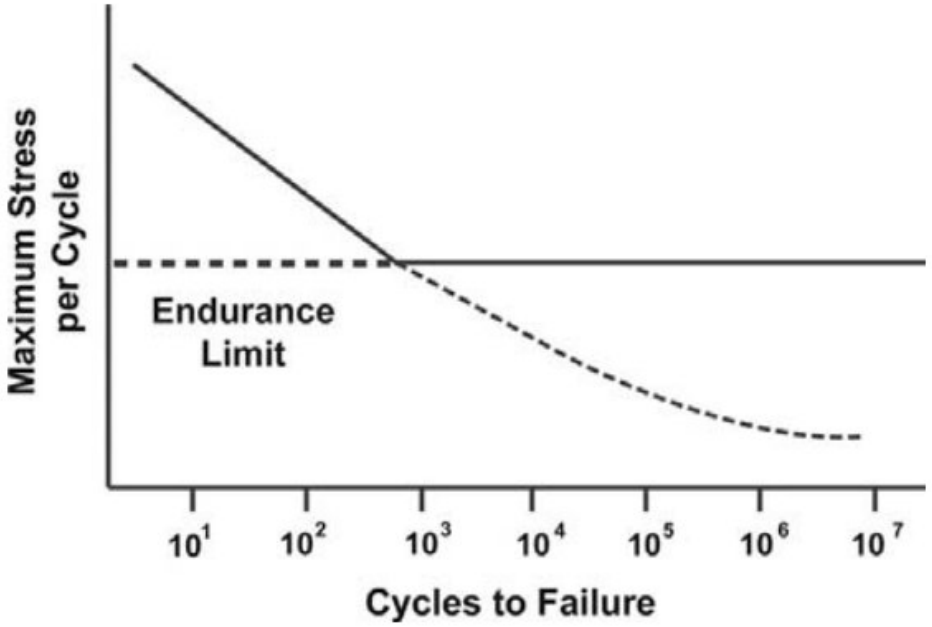


Figure 2.12

Examples of fatigue *S–N* curves. Solid lines show the behavior of a material with an endurance limit (below a certain load the material will not undergo fatigue failure). Dotted line shows typical behavior for a material which does not exhibit a fatigue limit.

Fatigue: Endurance limit

- Many materials have a minimum stress level below which fatigue failure does not occur
- This is called the endurance limit
- At this point, the S–N curve becomes horizontal

Important exception

- Some metals, such as aluminum and its alloys, do not have an endurance limit
- Their S–N curve keeps decreasing as the number of cycles increases
- Such materials should be avoided in applications with cyclic loading

Electrochemical properties

Corrosion

Corrosion : What is corrosion?

- Corrosion is the chemical deterioration of a material due to interaction with its environment
- This general definition can apply to many materials
- But here, the focus is on corrosion in metals

Main types of corrosion

- Corrosion is broadly divided into:
  - Wet corrosion → occurs in the presence of a liquid
  - Dry corrosion → occurs due to corrosive gases
- In biomaterials
- For biomaterials, wet corrosion is the important type
- This is because implants are placed in the fluid-rich in vivo environment

Why corrosion matters in implants

- Corrosion of implants is important because the body environment is complex
- Implants are exposed to many chemical substances
- The rate and severity of corrosion depend mainly on:
  - Chemical composition of the environment
  - Temperature
  - Stress level

Example 1: Iron in hydrochloric acid

- When iron reacts with **hydrochloric acid**, a strong corrosion reaction occurs
- **Hydrogen gas** is released

- Main idea
  - In corrosion, **oxidation and reduction happen at the same time**
  - They occur at **equal rates**
  - The final result is the **loss of iron**

The overall reaction is given in **Eq. (2.9)**
Partial reactions
Eq. (2.10) → anodic reaction / **oxidation**
Eq. (2.11) → cathodic reaction / **reduction**
Eq. (2.12) → formation of **ferrous chloride**

$$Fe + 2HCl \rightarrow FeCl_2 + H_2. \quad (2.9)$$

$$Fe \rightarrow Fe^{+2} + 2e \text{ (anodic reaction or oxidation)}, \quad (2.10)$$

$$2H^+ + 2e \rightarrow H_2 \text{ (cathodic reaction or reduction)}, \quad (2.11)$$

$$Fe^{+2} + 2Cl^- \rightarrow FeCl_2. \quad (2.12)$$

Example 2: Iron in water with dissolved oxygen

- Iron can also corrode in an **aqueous environment containing oxygen**
- The reactions are shown in:
  - **Eq. (2.13)** → oxidation
  - **Eq. (2.14)** → reduction
  - **Eq. (2.15)** → formation of **ferrous hydroxide**
- **Rust formation**
  - Ferrous hydroxide **$Fe(OH)_2$** is unstable
  - It is further oxidized to form **ferric hydroxide $Fe(OH)_3$**
  - This is commonly known as **rust**
  - This reaction is shown in **Eq. (2.16)**

$$2Fe \rightarrow 2Fe^{+2} + 4e \text{ (oxidation)}, \tag{2.13}$$

$$O_2 + 2H_2O + 4e \rightarrow 4OH^- \text{ (reduction)}, \tag{2.14}$$

$$2Fe^{+2} + 4OH^- \rightarrow 2Fe(OH)_2. \tag{2.15}$$

$$2Fe(OH)_2 + H_2O + \frac{1}{2}O_2 \rightarrow 2Fe(OH)_3. \tag{2.16}$$

Passivity

- What is passivity?
- Passivity is a special behavior seen in some metals
- Normally, as the oxidizing power of a solution increases, the corrosion rate also increases
- This first region is called the active zone
- Passive behavior
- In some metals, such as:
  - Iron
  - Chromium
  - Titanium
  - and their alloys
- Increasing oxidizing power beyond a critical point causes the corrosion rate to decrease
- This behavior is shown in Figure 2.13

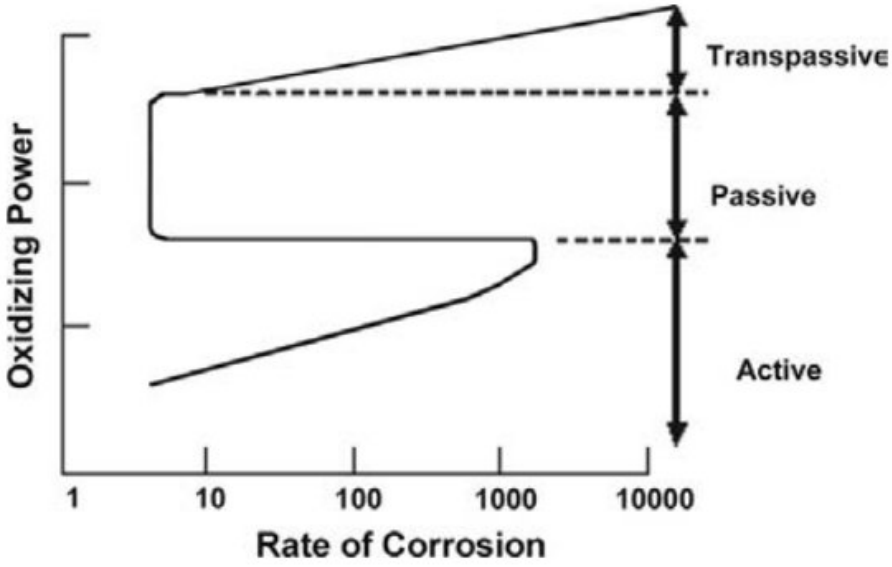


Figure 2.13

Corrosion behavior of an active–passive material as a function of electrode potential.

Passivity

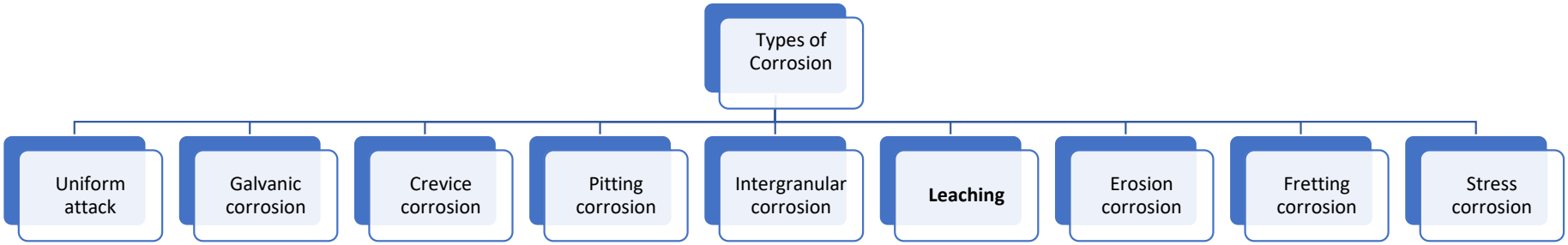
- Passive region
  - After the active zone, the metal enters a **stable passive region**
  - In this region, further increase in oxidizing power has **little or no effect** on corrosion rate
  - The metal behaves almost like an **inert noble metal**
- Transpassive region
  - If the oxidizing power increases even more beyond the passive region, the corrosion rate rises again
  - This region is called the **transpassive region**

Types of Corrosion

- Wet corrosion can occur in several forms
- Some forms are more important for implants than others
- But all forms should be considered in design



Types of Corrosion



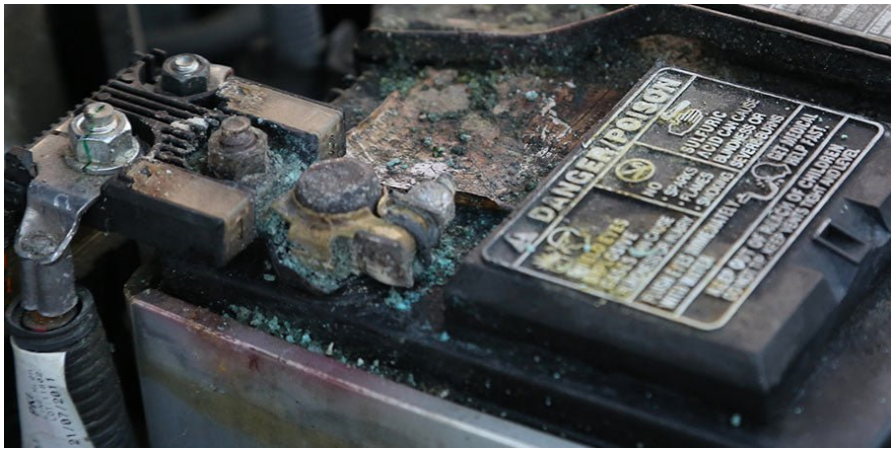
1. Uniform attack

- General surface corrosion over the whole metal.
- In biomedical use: can happen on implant surfaces in body fluids.



2. Galvanic corrosion

- Happens when two different metals are in contact in body fluid.
- Biomedical examples: modular hip joints, dental implants, plates and screws, intramedullary nail with wire.





3. Crevice corrosion

- Happens in narrow spaces where fluid gets trapped.
- Biomedical examples: cracks, modular joints, beneath screw heads.



4. Pitting corrosion

- Localized corrosion that forms small pits/holes.
- In implants, it can damage the surface and change its texture.



5. Intergranular corrosion

- Corrosion along grain boundaries of metals.
- In biomedical metals, it can reduce strength and in severe cases grains may separate.



6. Leaching

- Selective dissolution of one component from an alloy.
- Biomedical importance: especially important in nickel-based alloys because nickel may cause adverse biological response.

7. Erosion corrosion

- Corrosion accelerated by relative movement between metal and fluid.
- In biomedical context, movement can remove the protective layer and expose fresh metal.



8. Fretting corrosion

- A type of erosion corrosion at contact areas with vibration or slip.
- Biomedical examples: screws with fixation plates, rods, modular hip prostheses.



9. Stress corrosion

- Happens when tensile stress helps corrosion and crack growth.
- Biomedical example: a loaded implant in the body, especially under bending, may corrode more and crack.



Surface properties

Why surface properties matter

- The surface of a material is different from the bulk
- Surface atoms may have unsatisfied bonds
- So, the surface has higher energy than the bulk
- This extra energy is called surface energy
- Surfaces are highly reactive
- They may undergo:
  - Oxidation
  - Other chemical changes
- So, surface properties often differ from bulk properties

Why this is important in biomaterials

- The body interacts first with the surface, not the bulk
- After implantation, the surface is quickly coated by:
  - Proteins
  - Other molecules
- These surface coatings influence:
  - Cell response
  - Immune response
- So, controlling surface properties is very important for the long-term success of an implant

Examples of surface variation

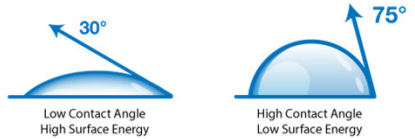
- Surface roughness may vary from smooth to rough
- Surface morphology may be:
  - Smooth
  - Stepped
  - Pockmarked
- The surface may contain different kinds of atoms
- Their arrangement may vary across the surface and in depth

Surface characterization

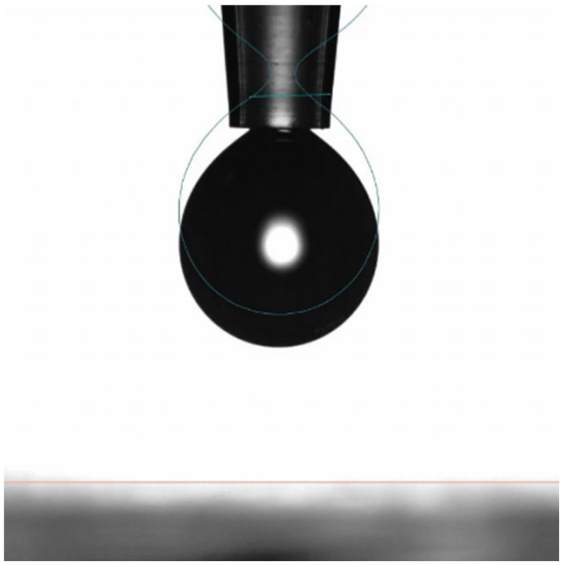
- Because the surface layer is very thin, surface testing needs:
  - Different methods
  - Higher sensitivity
- This section gives two examples of surface-property measurements:
  - Contact angle
  - Hardness

1. Contact Angle

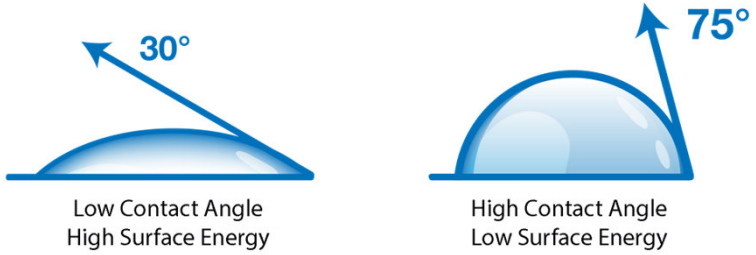
- When a liquid drop is placed on a surface, it may:
  - Spread out
  - Or remain as a droplet
- This depends on the balance between:
  - Cohesive force = attraction between liquid molecules
  - Adhesive force = attraction between the liquid and the surface
- Young's equation
- Surface energy can be related using Eq. (2.19)
- θ is the contact angle between the interfaces



$$\gamma_{sg} = \gamma_{sl} + \gamma_{lv} \cos\theta, \qquad (2.19)$$

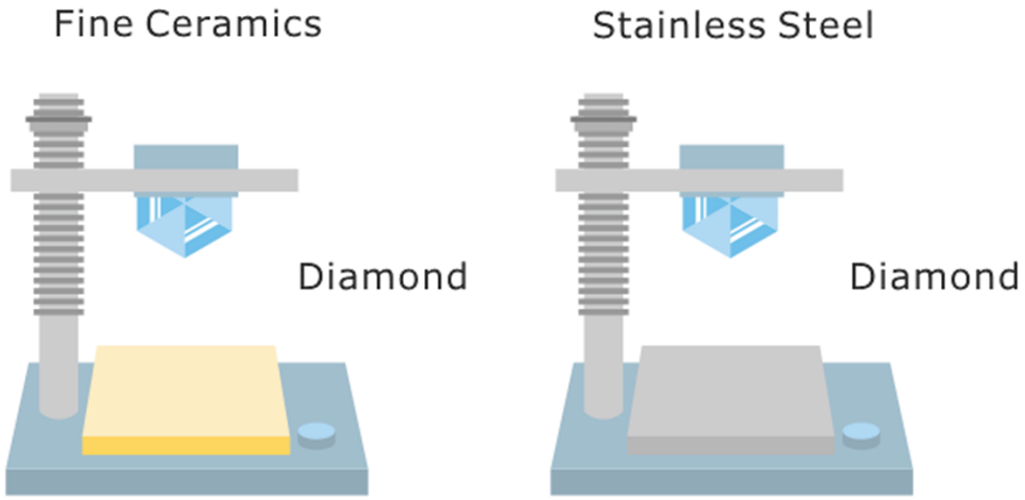


- Important cases
  - Complete wetting: θ = 0°
  - Complete non-wetting: θ = 180°
- Simple example
  - A water droplet behaves differently on a car surface before and after waxing

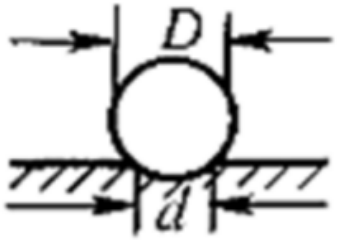
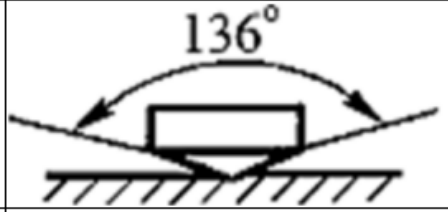
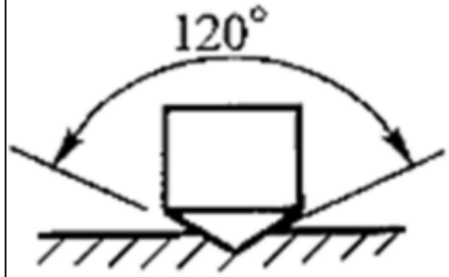
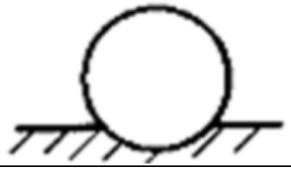


2. Hardness

- Hardness is the ability of a material to resist localized plastic deformation
  - It is especially important at the surface
  - Although it can be related to bulk properties like E and σy, it is better to measure it directly on the surface
  - Hardness is important when two materials will contact each other during use
- How hardness is measured
  - Usually measured by indentation
  - A standard indenter of known shape is pressed into the surface under a known load
  - Hardness is determined from:
    - Indentation depth
    - Or indentation area


Fine Ceramics
Diamond
Stainless Steel
Diamond

- Common hardness tests
  - Rockwell → uses spheres or cones; shallower indentation means higher hardness
  - Vickers → uses a square pyramidal indenter
  - Brinell → uses spheres; tests a larger area and is less affected by grain boundaries
  - Mohs → scratch test based on comparison with known minerals; mainly a relative test

Hardness Test Method	Indenter Shape	Indentation shape
Brinell Hardness（HB）		
Vickers Hardness（HV)		
Rockwell Hardness（HR)	 	 

Mohs Hardness Scale

Increasing Hardness

Mineral Name	Scale Number	Common Object
Diamond	10	
Corundum	9	Masonry Drill Bit (8.5)
Topaz	8	
Quartz	7	Steel Nail (6.5)
Orthoclase	6	
Apatite	5	Knife/Glass Plate (5.5)
Fluorite	4	Copper Penny (3.5)
Calcite	3	
Gypsum	2	Fingernail (2.5)
Talc	1	





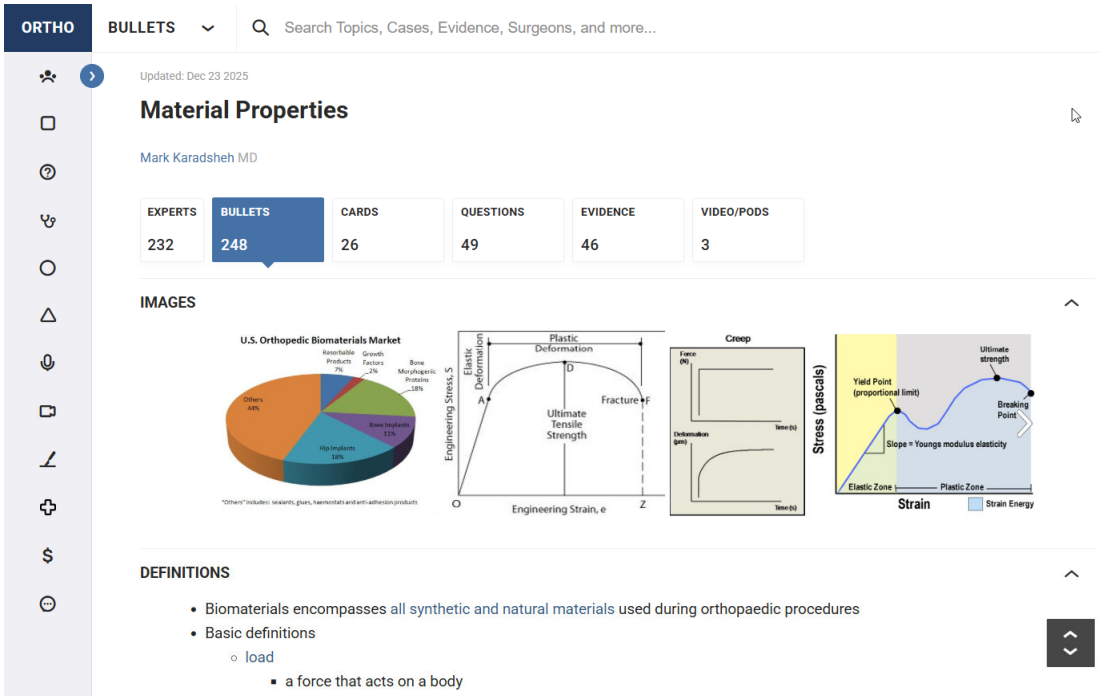






Material Properties

- https://www.orthobullets.com/basic-science/9062/material-properties



SO→

- Material behavior in implants depends on:
  - Mechanical properties
  - Electrochemical properties
  - Surface properties
- Standard tests help predict behavior and reduce risk of failure
- For commercial biomaterials, these properties are often given in specification sheets
- For self-produced materials, standard testing is needed to fully characterize them

Think with me 🤔🤔🤔

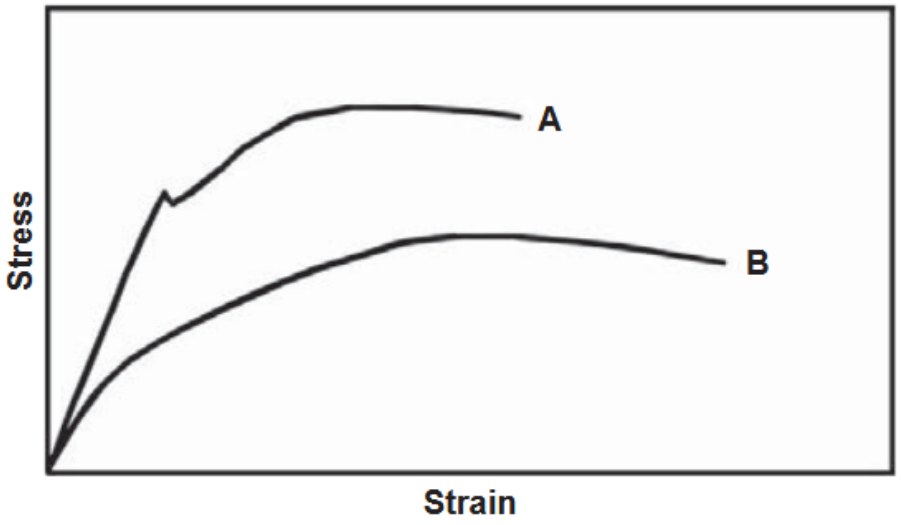
The figure below shows the stress–strain curves for two different materials A and B. Which of these materials:

(a) is stronger,

(b) is more ductile,

(c) will absorb more energy prior to failure?

(d) Based on the stress–strain curves, what is the likely general material type of A and B?




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Thank you